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Union negotiations and corporate policy

A study of labor concessions in the domestic steel industry during the 1980s*

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This paper studies managerial compensation, financial reporting, and dividend policies of the seven major domestic steel producers during requests for union concessions. Substantial layoffs, reported losses, and sacrifices by nonunion stakeholders buttressed managers' case for union concessions. From 1980 to 1988, sample firms reduced their work force by about 300,000 (almost two-thirds) and annual wage payments from $16.1 to $8.6 billion. Reported income is lower during union negotiations, controlling for cash flows, as is managerial pay, with average CEO salary plus bonus declining 18%. Dividend reductions and white-collar pay cuts are substantial, pervasive, and clustered during union negotiations.

1. Introduction

Most studies of corporate policy focus on changes in one decision – in dividend payments, managerial compensation, or accounting choice – as determined by corporate performance. Largely unaddressed is the extent to which these decisions *interact* to reflect a consistent overall corporate strategy. For example, are changes in dividend policy linked to changes in managerial compensation? If so, do these corporate policy decisions reflect

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strategic considerations, for example, information costs characteristic of the firm's political environment, in addition to technological factors? Investigation of these issues is difficult in large samples, since it requires analysis of a great deal of firm- and industry-specific data. Hence we focus on firms in one industry, the domestic integrated steel producers. These firms' generous labor practices, established in prior negotiations with a powerful union, came under strong pressure in the 1980s as a result of changes in the competitive environment that also created incentives for managers to adjust corporate policy.

During the 1980s, U.S. steel producers faced increased competition from subsidized imports and nonunionized mini-mills, the technological obsolescence of aging plants, and a decline in steel demand due to a worldwide recession. The resultant overcapacity led to years of massive losses – $3.2 billion in 1982, the first industry loss since the Great Depression, followed by $9.5 billion in the 1983–1986 period for the seven major integrated producers.[1] In 1982, domestic steel production fell to its lowest level since 1946, with capacity utilization at 48.1% for the major producers. (Steel production entails large fixed costs, with breakeven then estimated at 80–85% of capacity, in part because of high labor costs.) By 1982, the average domestic steelworker's total wages and benefits exceeded by more than 60% those paid the more productive Japanese. Moreover, this competitive disadvantage was itself costly to remove, since existing labor agreements with the United Steelworkers of America (USW) granted severance benefits estimated as high as $60,000–$70,000 per steelworker.

Managers of domestic steel producers faced the difficult task of negotiating concessionary union agreements that would permit their firms to reduce capacity and modernize production in response to increased competition and declining demand. This paper studies adjustments in managerial compensation, financial reporting, and dividend policy made by managers of the seven largest domestic steel producers during the period in which they negotiated substantial givebacks by unionized labor. These difficult labor negotiations led to the breakup of the multi-employer bargaining group that had bargained for the domestic steel industry for 30 years, and resulted in temporary wage reductions and other union concessions in both 1983 and 1986–1987.

The 1980s restructuring of the domestic steel industry had truly dramatic consequences for steel producers' labor costs, employment, and productive capacity. The aggregate wage bill for the seven major producers, which totalled about $16.1 billion in 1980, declined a striking 46.6% to around $8.6

[1]These firms, which make up our sample, had approximately two-thirds of the domestic steel industry's productive capacity during the 1980s. The firms are U.S. Steel Corporation (now USX), LTV Corporation, Bethlehem Steel Corporation, Armco Inc., National Steel Corporation (later National Intergroup), Republic Steel Corporation (now owned by LTV), and Inland Steel Company.

billion in 1988. This substantial reduction in total labor costs was largely achieved through radical employment cuts. Steelworker layoffs at the seven firms were pervasive and resulted in a net reduction of almost two-thirds of the work force, for a total loss of approximately 300,000 jobs over the decade. This radical employment decline reflects sample firms' decisions to downsize operations that reduced their productive capacity by 38.2% from 1980 to 1988.

To obtain labor concessions, managers must make a credible case that current economic conditions pose a serious threat to the firm's competitive viability. Perhaps the most convincing evidence is that managers actually shut down some operations and effected large-scale layoffs. Substantial losses provide corroboratory evidence. We find that sample firms' reported net income is significantly lower during union negotiations, controlling for cash flows, than in nonnegotiation years. The losses reported in negotiation years are largely driven by one-time special charges that reflect managers' real restructuring decisions, but whose timing is discretionary across negotiation and nonnegotiation years. These charges are commonly taken late in the fourth quarter, when managers announce that they *expect* to shut down operations (or not reopen previously idled plants), announcements that could readily be deferred into the next fiscal year. It thus appears that managers of domestic steel firms deliberately reported larger losses during union negotiations to help their case for wage and other concessions from the USW.

We also find a consistent pattern in other managerial decisions that strengthens the case that serious financial problems require major concessions from unionized labor. Specifically, managers' requests for union concessions were bolstered by substantial contemporaneous sacrifices by stakeholders other than union workers. We find pervasive pay cuts for white-collar workers, the timing of which seems designed to convince unionized labor to follow suit. Dividend reductions are also pervasive and are concentrated during union negotiations. Dividends were eliminated by six firms (although the median firm had paid dividends for 46 continuous years) and cut substantially by the seventh. Cash conserved via dividend reductions aggregates to several hundred million dollars per year which, while surely not trivial, is well below the billions of dollars in annual savings accomplished via layoffs and wage reductions.

Top management's compensation was cut at all seven firms, with especially large reductions in union negotiation years, controlling for corporate performance. Chief executive officers (CEOs) experienced an average 18.1% decline in salary plus bonus in negotiation years (versus an average 19.5% increase in other years). For the five highest paid officers, salary plus bonus fell by an average 15.4% in union negotiation years (versus an average 16.0% increase in other years). Although these data indicate that top managers took large percentage pay cuts during union negotiations, the resulting cash flow

savings are quite small compared to the annual labor cost savings we document. Top management's financial sacrifices thus appear to be symbolic actions which indicate that they too will bear personally significant costs during the firm's financial difficulties. In fact, top management often stresses its own sacrifices when attempting to gain concessions from unionized labor and other stakeholders.

Overall, our results conform closely to the political view of the corporate contracting process outlined in DeAngelo and DeAngelo (1990) on dividend policy and in Jensen and Murphy (1990a, b) on managerial compensation. DeAngelo and DeAngelo suggest that dividend reductions reflect managers' desire to reconcile the interests of various claimants during periods of financial distress. Jensen and Murphy suggest that it is difficult to reward top management adequately for excellent performance because such rewards would engender significant protests from workers. The result is a contracting environment in which the payoffs to top management, unionized labor, and others are implicitly linked. Thus, when seeking union concessions, managers have incentives to induce and publicize sacrifices by all stakeholders, so that corporate policies are determined by strategic contracting considerations as well as by fundamental factors such as the firm's technological capabilities, product market pressures, etc.

Section 2 of the paper briefly describes the competitive environment in the steel industry during the 1980s. Section 3 documents sample firms' reductions in employment and labor costs, and describes the timing of union negotiations. Section 4 reports sample firms' earnings and cash flow performance, and relates earnings performance and the timing of noncash writeoffs to the occurrence of union negotiations. Section 5 describes wage reductions for managerial and salaried personnel, and relates their timing to that of union negotiations. Section 6 relates the timing and stock price impact of dividend reductions to union negotiations. Section 7 concludes with a brief summary.

2. The domestic steel industry in the early 1980s[2]

In the early 1980s, the U.S. steel industry consisted of seven major integrated producers that accounted for about two-thirds of domestic capacity, some smaller producers, and a number of domestic nonunionized minimills that reproduce steel from scrap. Steel is a primary input in the production of automobiles, industrial machinery and equipment for the construction, railroad, and shipbuilding industries, motors of all kinds, build-

[2]This section's discussion of overall conditions in the domestic steel industry is based on various issues of *Standard & Poor's Industry Surveys*, articles in the *Wall Street Journal* (*WSJ*), and Hoerr (1988). The data in this and the following sections are compiled from corporate annual reports, forms 10K, proxy statements, the Center for Research in Security Prices and Compustat tapes, *Moody's Manuals*, the *American Iron and Steel Institute 1988 Annual Statistical Report, Standard & Poor's Industry Surveys*, and the *WSJ*.

ings and bridges, office furniture, and appliances. Since most steel output is used in the production of capital goods, demand is highly cyclical. However, the costs of steel production – primarily production equipment and unionized labor – are relatively fixed. Thus steel production is unprofitable at relatively low levels of output. At the beginning of the decade, the breakeven point for steel production was estimated at 80–85% of productive capacity (see, for example, *Standard & Poor's Industry Surveys*, April 28, 1983, p. S15).

Before the worldwide recession of the early 1980s, a number of factors had already reduced the demand for domestic steel. First, the oil crisis of the early 1970s had encouraged automobile manufacturers to switch production toward lighter, more fuel-efficient cars that used more aluminum and plastic materials. (Automobile manufacturers, which accounted for 21.4% of domestic steel shipments in 1973, accounted for only 14.9% by 1980.) Second, imported steel had captured an increasing portion of domestic consumption – rising from 14.1% in 1976 to an eventual high of 26.2% in 1984 (*Standard & Poor's Industry Surveys*, February 3, 1986, p. S26). Because of the importance of steel production to basic industry, many foreign governments subsidized it, and the rising value of the dollar also helped foreign producers. Finally, integrated producers faced increased competition from domestic mini-mills that were able to reproduce steel from scrap at a cost advantage over new steel, both because of low prices for scrap and because they employed nonunionized labor.

Substantial costs of exiting steel production exacerbated the resultant overcapacity problem. These costs were due primarily to existing labor agreements with the USW, and included severance pay, pension and health care benefits, unemployment insurance, and other supplemental unemployment benefits. (They also included federally mandated environmental reclamation costs.) The total employment costs of permanently closing a steel production facility were estimated to be as high as \$60,000–\$70,000 per employee (see, for example, 'Bethlehem's Plan to Slash Capacity Shows How Fast U.S. Steel Industry Is Shrinking', *The Wall Street Journal*, December 29, 1982). The problem, as Fruhan (1985, p. 133) has pointed out, is that 'if a steel company wants to close a plant, it becomes obligated to pay unreserved and unfunded pension and health care costs that generally far exceed the liquidation value of the plant and related working capital'.

Managers of domestic steel firms thus found themselves in a difficult dilemma – although capacity reductions were costly, so was maintaining production at current labor costs. By 1982, the average U.S. steelworker's total wages and benefits had reached nearly \$24 per hour. This figure exceeds by more than 60% what Japanese steelworkers earned, although the Japanese were estimated to be twice as productive (see, for example, *Standard & Poor's Industry Surveys*, April 28, 1983, p. S15). It is also high relative to the historical premium paid steelworkers over employees of domestic manufac-

turing firms. We document below that, by 1982, the average hourly total compensation of domestic steelworkers exceeded that paid other manufacturing employees by 73%, up from a premium of 24% in 1964.[3] Although employment in a steel plant unquestionably merits a premium because of the dangers and difficulties of the work, the substantial growth in the compensation premium paid steelworkers implausibly reflects a tripling of job risk from 1964 to 1982.

Fruhan (1985) attributes the substantial premium increase to steelworkers' generous wage and fringe benefit increases negotiated under the Experimental Negotiation Agreement (ENA). The ENA, enacted in 1973, provided for a minimum 3% annual wage increase and automatic cost-of-living adjustments in exchange for the prohibition of industry-wide strikes. From 1956 to 1980, the major domestic integrated steel firms had bargained collectively for the industry, with negotiations taking place at three-year intervals. Despite signs of increasing trouble in the industry, the 1980 negotiations (opened in February and settled in April) granted a wage hike estimated to increase labor costs by 37% over the next three years (*Standard & Poor's Industry Surveys*, October 30, 1980, p. S49).

Our sample consists of the seven major domestic integrated steel producers that comprised the 1982 Basic Industry Bargaining Group.[4] Table 1 lists the seven firms, and reports their *Fortune 500* rank in 1979 (column 1) and in 1989 (column 2). Although all seven firms placed in the 1979 *Fortune 100*, only three made the list in 1989, and every firm declined in rank. LTV declined from 31 to 72 despite its 1984 acquisition of Republic Steel (LTV cum Republic filed for Chapter 11 bankruptcy in 1986). Although U.S. Steel's rank fell only from 14 to 19, its continued high placement is due in part to the acquisitions of Marathon Oil in 1982 and Texas Oil & Gas in 1986, which eventually led *Fortune* to reclassify it as an oil firm.

Sample firms' decline in *Fortune* rank is accompanied by a substantial decline in productive capacity and sales. Productive capacity for all domestic steel producers decreased by 27.1%, from 153.7 million tons of raw steel in 1980 to 112 million tons in 1988 (*Standard & Poor's Industry Surveys*, March 9, 1989, p. S5). For our sample firms, productive capacity declined by 38.2%, from 107.5 million tons in 1980 to 66.4 million tons in 1988, and total sales

[3]This calculation is adopted from Fruhan (1985, exhibit V). Fruhan reports that, relative to workers in manufacturing, steelworkers received a compensation premium of 92% in 1982, up from 33% in 1964. The difference between our estimates and those of Fruhan is attributable to use of different manufacturing wage series generated by the Bureau of Labor Statistics. Both estimates indicate large increases over the 1960s and 1970s in steelworkers' compensation premium.

[4]An eighth firm, Wheeling-Pittsburgh Steel, which later filed for bankruptcy protection, was expelled from the group in 1981 for attempting to negotiate a separate wage agreement with the USW. It was ultimately replaced by Allegheny Ludlum, a specialty steel firm, which we do not include in the current sample because it is not a major integrated producer.

Table 1

Fortune 500 rank of the seven major domestic steel firms in the 1982 basic industry bargaining group.

	1979 rank	1989 rank
U.S. Steel Corporation (USX)	14	19[a]
LTV Corp. (Jones & Laughlin)	31	72[b]
Bethlehem Steel Corp.	38	95
Armco Inc.	58	178
National Steel Corp.	76	171[c]
Republic Steel Corp.	85	n/a
Inland Steel Company	95	117

[a]Ranking of USX in 1989 includes acquisitions of Marathon Oil in 1982 and Texas Oil & Gas in 1986, which led Fortune to classify USX as an oil firm in 1989.

[b]Ranking of LTV in 1989 reflects the firm's 1984 acquisition of Republic Steel. LTV filed for bankruptcy in 1986.

[c]Ranking of National Steel in 1989 (then National Intergroup) does not include its former plant, Weirton Steel, which was then employee-owned and itself ranked 288 on Fortune's 1989 list.

fell from $43.6 billion to $40.3 billion (a 7.6% decline). The latter figure, however, does not take into account the fact that some companies diversified into nonsteel operations during this period, which tends to inflate the end-of-period sales figures.[5] As a percent of total revenues, revenues from steel production fell from 70.2% in 1980 to 57.5% in 1988. Accordingly, a better measure of the steel industry's economic decline in the 1980s is the change in the seven firms' aggregate revenues from steel production. This figure fell 24.4% between 1980 and 1988, from $30.7 billion to $23.2 billion.

Fig. 1 shows the stock price performance of the seven major steel firms and the CRSP value-weighted market index over the 1980–1988 period. Steel firms' average stock price performance closely tracks that of the market through the recession of the early 1980s. After 1984, steel firms consistently underperform the market. For example, at the end of 1983, the compounded cumulative average stock return is 69.7% for the steel firms, versus 52.8% for the market index. By the end of 1988, the cumulative stock return is 25.3% for the steel firms, versus 157.3% for the market index.[6] Thus, steel produc-

[5]U.S. Steel's acquisition of Marathon Oil and Texas Oil & Gas clearly had this effect. Marathon's 1981 sales were $9.22 billion and Texas Oil & Gas's 1985 sales were $1.68 billion. These figures suggest the 1988 aggregate revenue of $40.3 billion is overstated by about $10.9 billion. The adjusted sales figure translates to a 32.6% decline from 1980 to 1988.

[6]The stock price performance of individual steel firms (not reported here) is similar to the sample average reported in fig. 1, with two exceptions. LTV, which filed for Chapter 11 bankruptcy protection in 1986, outperformed the market until early 1985, with severe deterioration in its performance thereafter (LTV's cumulative return was −68.9% at the end of 1988). U.S. Steel's stock price performance closely tracks the market index over the 1980–1988 period, with a cumulative return of 170.9% over the entire period.

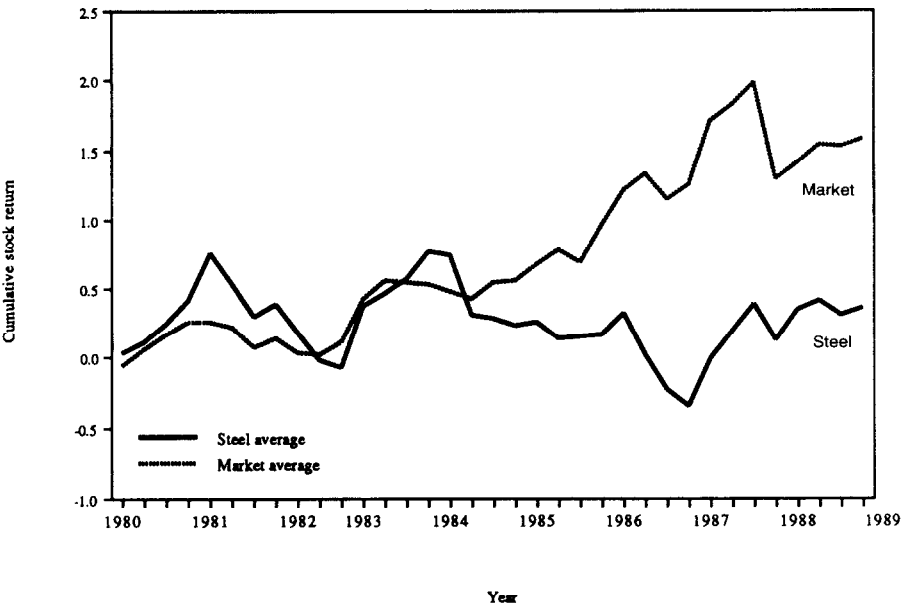


Fig. 1. Average stock price performance of the seven major integrated steel producers versus the value-weighted market index, 1980–1988.

'Steel average' refers to the compound cumulative average raw (dividends and capital gains inclusive) rate of return for the common stocks of the seven major domestic integrated steel producers. Republic Steel is directly included in the average only until its 1984 merger with LTV (and thereafter is included via LTV's stock return). 'Market average' refers to the cumulative rate of return on the CRSP value-weighted market index. All data are drawn from CRSP.

ers fared no worse than the market through 1984, but have lost substantial ground since. Overall, changes in *Fortune* rank, productive capacity, and sales uniformly indicate that the seven major domestic steel producers suffered a material economic decline during the 1980–1988 period.

3. The timing and impact of labor concessions

Although the 1980 contract with the USW did not expire until August 1983, negotiations were opened early (in June 1982) to accommodate the steel industry's request to bargain for wage concessions. For these talks, the seven major producers chose not to invoke the no-strike provisions of the ENA (because they were unwilling to grant the automatic wage increases it required). They held on-again, off-again negotiations with the union during

1982, failing to reach an agreement first in July and again in November. Ultimately they reached a three and one-half year agreement that went into effect in March 1983. This agreement provided for a 9% reduction in hourly pay, and reductions in vacation pay, cost-of-living adjustments, and other benefits (with gradual reinstatement over the life of the contract) that were estimated to reduce 1983 aggregate labor costs by some $1.5 to $2 billion.

Dissent over how to negotiate the requisite union concessions caused the steel industry bargaining group to collapse before the expiration of the 1983 contract. As a result, the major steel firms negotiated separate labor agreements during 1986–1987 (for the first time in 30 years). LTV, Bethlehem Steel, National Intergroup, and Inland Steel had all negotiated wage agreements by the contract expiration date of July 31, 1986. LTV filed for bankruptcy in July, however, and soon thereafter began an attempt to negotiate a new contract (ratified in August 1987). Armco negotiated a three-month extension, and had reached an agreement with the USW by November 1986. U.S. Steel was struck by the USW on August 1, 1986; when a new contract was ratified in February 1987, the strike had lasted longer than any other in the industry's history. Overall, 1986–1987 wage agreements provided for temporary wage and benefit concessions in return for an increase in profit sharing and job security.

Appendix A provides a more detailed chronology of union negotiations in the steel industry between 1980 and 1988. It indicates that industry requests for wage concessions were made during two periods: June 1982 to March 1983 for all firms, and January 1986 to August 1987, with exact dates differing for individual firms. In sections that follow, we relate changes in reported earnings and in managerial compensation to the presence or absence of union negotiations. All seven firms reported earnings on a calendar year basis (until National Steel switched to a March 31 year-end in 1986).

Accordingly, earnings released during requests for labor concessions are 1982 earnings for all seven firms, 1985 earnings for five firms (Republic had merged with LTV, reducing the sample to six firms, and U.S. Steel did not open talks until June 1986), and 1986 earnings for two firms (LTV and U.S. Steel had not settled by the 1986 earnings release date). Managerial compensation figures are taken to be those related to the same calendar years. For ease of discussion, we shall refer to these observations for 1982, 1985, and 1986 as 'union negotiation years'. (Strictly speaking, these observations refer to financial data for 1982, 1985, and 1986 that are relevant for union negotiations held respectively in 1982–1983 and in 1986–1987.)

As a result of reductions in the work force, wage concessions, and productivity improvements, the seven major domestic steel producers experienced a sharp decline in total annual labor costs from 1980 to 1988. (Because sample firms do not systematically report wage expense and employment data for union versus nonunion employees or steel versus nonsteel employees, our

data necessarily refer to the pooled data that are publicly available.[7]) Sample firms' aggregate labor costs declined from about $16.1 billion in 1980 to around $8.6 billion in 1988, a decline of $7.5 billion. Alternatively, we can estimate the decline in steel-related labor costs by taking 1988 total labor costs times the ratio of steel revenue to total revenue in that year, minus the same calculation for 1980. The latter approach yields an estimate of the aggregate dollar steel-related labor cost savings of $6.4 billion per year.

Both estimation approaches imply multi-billion dollar annual labor cost savings which, of course, loom even larger when the annual flow amounts are capitalized to obtain the present value of the savings. Although our estimates necessarily include some imprecision (see footnote 7) and do not precisely separate steel-related union wages from other labor costs, their size leaves little doubt that (i) potential cost savings from renegotiating labor contracts were material at the beginning of the decade, and (ii) sample firms did, in fact, achieve material labor cost savings over the decade.

In 1980, the average sample firm reported labor costs of 36.1% of sales (median, 35.5%).[8] These costs range from 30.0% of sales for Armco to 44.1% for Bethlehem Steel. By 1984, average labor costs had fallen to 30.8% of sales (median, 32.7%), and range from 17.4% for U.S. Steel to 40.2% for Bethlehem (U.S. Steel's reductions partially reflect the firm's 1982 acquisition of Marathon Oil). By 1988, average labor costs had fallen to 24.6% of sales (median, 25.8%) in our sample. Over the 1980–1988 period, then, sample firms reported an average 31.9% decline in labor costs as a percent of sales (median decline, 27.3%).

This secular decline in labor costs was accomplished to a large degree through radical reductions in the number of workers, which fell by about 300,000 (two-thirds of the work force) over the decade. Aggregate employment by the seven major steel producers declined by 62.9%, from 512,941 in 1979 to 190,238 in 1988, even though U.S. Steel acquired Marathon Oil and Texas Oil & Gas in the interim.[9] Thus, over the decade, an estimated 322,703

[7]To estimate the seven firms' aggregate dollar labor costs, we approximate labor costs for firms that did not report them by taking the product of the firm's sales times the average labor cost/sales ratio for firms that did report labor costs for the year in question. [Labor cost data are not reported by LTV (1980–1988), Republic (1983, the last year before its merger with LTV), and National Steel (1983–1988, coinciding with the sale of Weirton Steel).] Some evidence that this procedure yields plausible labor cost estimates is that its estimates for National Steel for 1980–1982 fall within −8.4%, −6.9%, and 0.7%, respectively, of the actual labor costs reported by the firm.

[8]These averages are for all firms that reported labor costs (see footnote 7).

[9]Acquisition of these two firms probably inflates the 1989 employment figures by about 20,000. (Marathon employed 16,168 people in 1981 and Texas Oil & Gas employed 3,629 in 1985.) Additionally, the 1988 employment figures are probably low by about 10,000, the approximate 1980 employment of Weirton Steel (a National Steel subsidiary that was sold to employees in 1982). It makes little difference if we adjust the employment figures in the text for these transactions – the data unquestionably indicate that domestic steel firms reduced employment by massive amounts during the 1980s.

Table 2

Total employment cost per hour for steel and manufacturing firms, 1964–1988.

Total employment cost per hour includes wages and employer expenditures on fringe benefits. Manufacturing total employment costs were obtained from the Office of Productivity of the Bureau of Labor Statistics. Steel data are drawn from the *American Iron and Steel Institute 1988 Annual Statistical Report* and Fruhan (1985).

Year	Steel	Manufacturing	Premium (%)
1988	$24.65	$16.86	46%
1987	23.71	16.31	45
1986	23.24	15.94	46
1985	22.81	15.25	50
1984	21.30	14.54	46
1983	22.21	14.08	58
1982	23.78	13.72	73
1981	20.16	12.60	60
1980	18.45	11.49	61
1979	15.92	10.27	55
1978	14.30	9.36	53
1977	13.04	8.63	51
1976	11.74	7.94	48
1975	10.59	7.33	44
1974	9.08	6.55	39
1973	7.68	5.89	30
1972	7.08	5.47	29
1971	6.26	5.20	20
1970	5.68	4.91	16
1969	5.38	4.58	17
1968	5.03	4.26	18
1967	4.76	3.96	20
1966	4.63	3.76	23
1965	4.48	3.59	25
1964	4.36	3.52	24

jobs were lost at the seven sample firms. For the entire steel industry, the number of steelworkers declined by 284,283 (or 62.7%), from 453,181 in 1979 to 168,898 in 1988 (*American Iron and Steel Institute 1988 Annual Statistical Report*).[10]

The data in table 2 suggest that steelworker wage rates were dampened significantly following the union negotiations studied here, but were not reduced to the levels that prevailed in the 1960s and early 1970s. For each year from 1964 to 1988, the table presents total employment costs per hour for steel and manufacturing firms and the implied hourly compensation premium paid to steelworkers. The data show that steelworkers' compensation premium increased substantially over the 1960s and 1970s which, as

[10]The number of employees for our sample firms exceeds that for the steel industry as a whole because our firms have nonsteel operations (and because they comprise a large portion of the industry). As noted above, data limitations preclude separate calculation of the number of steelworkers for our sample firms.

discussed earlier, contributed to the financial difficulties steel firms faced in the 1980s. As of 1980, steelworkers' average hourly compensation represented a 61% premium over the compensation of manufacturing workers (up from 24% in 1964 and, even more dramatically, from 16% as recently as 1970). After the union negotiations of the early to mid-1980s, steelworkers' hourly compensation premium shrank substantially, and leveled out to the 45–46% range by 1986–1988.

Had steel firms paid in 1988 the 61% compensation premium they paid in 1980, the average steelworker would have received 10.3% greater compensation. Accordingly, sample firms' 1988 total wage bill of $8.6 billion would have been $9.5 billion, or almost $1 billion higher had steelworkers' premium not been reduced from its 1980 level. (This estimate assumes that the level of employment is not reduced from its actual 1988 level in response to the hypothetical higher wage rate.) Had sample firms maintained employment at the 1980 level, their 1988 aggregate wage bill would have been higher by many billions of dollars, since they laid off almost two-thirds of their workforce over the period. Thus, while the cost savings from layoffs and dampened wage payments were each substantial in total dollar terms, layoffs were responsible for the bulk of the annual labor cost savings achieved by sample firms.

Appendix B provides a detailed description of the pervasive, large-scale steelworker layoffs and plant closures at sample firms during the 1980–1988 period. It indicates, for example, that Bethlehem Steel announced plans in 1982 to permanently shrink its steelmaking capacity by 20%, and to eliminate 10,000 jobs. Republic Steel announced layoffs in the same year that totaled 14,000 workers (more than one-third of its work force) and plans to shut down operations at several plants. In 1984, Inland announced further layoffs (beyond the 24% of its work force already laid off), and in 1985 announced plans to shut down operations representing 30% of productive capacity. The appendix documents numerous layoff announcements by each of these companies and for the remaining four sample firms.

Although layoffs and plant closings were announced throughout the early to mid-1980s, the union negotiation year of 1982 saw especially large employment reductions by sample firms. For all seven firms, appendix B indicates that more employees were let go in 1982 than in any other year between 1980 and 1988. It seems reasonable to believe that the large employment reductions in 1982 primarily reflect the intense economic pressure to reduce capacity that sample firms were under at the time. But it is also clear that managers used threatened and actual layoffs and plant closings to back up their claims that labor concessions were essential. For example, after U.S. Steel failed in 1983 to obtain additional union concessions at selected plants, the firm announced it would close almost all of the marginal operations in question, reduce steel capacity by 17%, and eliminate 15,400 jobs (see the

appendix A entry for December 1983 and the appendix B entry for U.S. Steel in 1983).

The threat of massive layoffs is a credible means of obtaining union concessions because the entire steel industry exhibited visible evidence of financial difficulty at the time. Should the USW fail to respond with wage concessions, their members stand to lose not only the value of any firm-specific human capital, but also the value of any industry-specific capital. This fact creates an incentive for managers of domestic steel producers to 'pull together' and jointly paint a picture of extreme financial distress to the union. Of course, managers of individual firms have incentives to create an even bleaker picture if they can thereby negotiate differential givebacks that provide an advantage over competitors – witness the bankruptcy filings of LTV and Wheeling-Pittsburgh Steel and the latter firm's 1981 expulsion from the multi-employer bargaining group.

4. Financial reporting strategy

One means managers can use to paint a bleak financial picture is to deliberately understate reported earnings. Although union leaders may be more financially sophisticated than the rank-and-file (and therefore better able to 'see through' managers' accounting choices), they may themselves truly believe the firm's condition merits concessions. Hence, low reported earnings may help union leaders convince the rank-and-file to accept contract terms that workers would ordinarily view as onerous. In any case, an empirical finding that earnings are more negative during union negotiations than would be predicted by sample firms' contemporaneous cash flows would provide evidence that managers used their firm's financial reporting strategy to encourage labor givebacks.[11]

Fig. 2 describes sample firms' aggregate annual operating cash flows, 'bottom line' net income, and operating income from steel production over the 1980–1988 sample period.[12] Although aggregate cash flows are positive throughout the entire period, aggregate net income is negative in each of the

[11]Another reason managers might depress earnings is to convince the government that their firms are sufficiently troubled to merit import relief [Jones (1990)]. The major steel producers, however, were engaged in seemingly continuous lobbying for import relief during the 1980–1988 period (according to annual report disclosures). Hence it is not possible to isolate particular years in which managers had incentives to depress earnings to aid their lobbying efforts.

[12]Operating cash flows are collected from annual reports. For firm-years for which cash flow or cash basis funds statements are presented, operating cash flows are taken directly from the statements. For firm-years for which working-capital basis funds statements are presented, operating cash flows are calculated conventionally as working capital from operations adjusted for changes in current balance sheet accounts considered to be operating [see Bowen, Burgstahler, and Daley (1986)]. Both approaches yield the conceptually identical cash flow number. Operating income from steel production is as reported in the footnotes to the financial statements, adjusted for steel-related unusual income items (normally considered operating items).

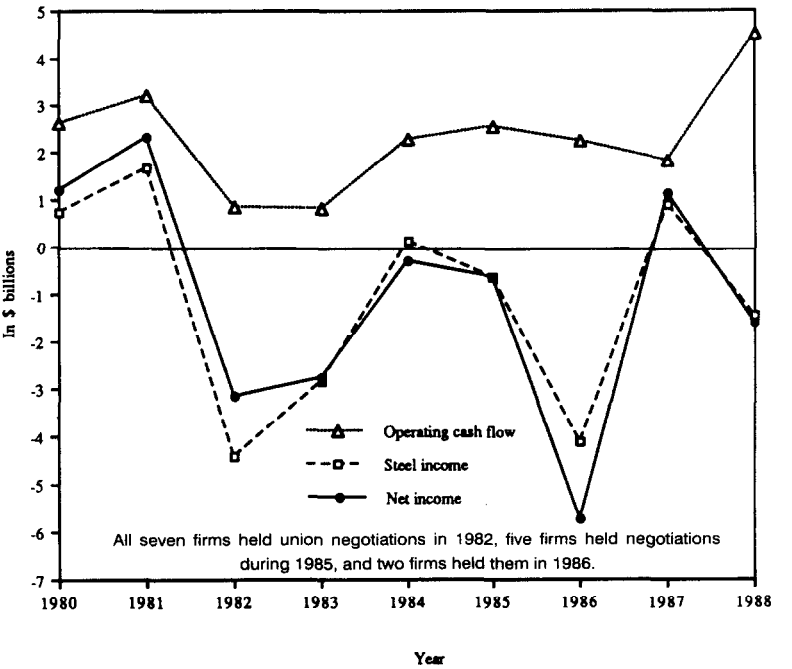


Fig. 2. Aggregate steel income, net income, and operating cash flow for the seven major domestic steel producers, 1980–1988.

Steel income is operating income from steel production plus the effect of steel-related unusual income items, such as provisions for the costs of plant closings. Net income is total 'bottom line' earnings and includes income from both steel-related and non-steel-related activities. 'Operating cash flow' is as reported by the firm in the statement of cash flow. If a statement of changes in financial position is presented, operating cash flow is taken from the statement or calculated as working capital from operations adjusted for changes in noncash current operating assets and liabilities. All seven firms held union negotiations in 1982, five firms held negotiations during 1985, and two firms held them in 1986.

years 1982 through 1986, and again in 1988. Thus fig. 2 reveals that cash flows systematically exceed earnings, and by substantial amounts in some years. Fig. 2 also shows that aggregate net income and steel income are quite closely related. They differ in sign only in 1984 and, as in all other years, are still very close in magnitude for that year. This fact suggests that net income closely approximates income from steel production for our sample firms.

The only year in which financial data are relevant for union negotiations by *all* sample firms is 1982, at which time the aggregate reported loss is $3.2 billion.[13] 1985 data are relevant for the five of six firms that held union negotiations in 1986. In 1985, all six firms reported an aggregate loss of $0.7

[13]As noted earlier, the financial data for 1982 are relevant for the union negotiations that ended in 1983. Similarly, the financial data for 1985 and 1986 discussed later in this paragraph are relevant for the 1986–1987 union negotiations.

Table 3

Parameter estimates (t-statistics) for the pooled regressions of earnings changes on changes in cash flow and presence of union negotiations for the seven major domestic steel producers, 1980–1988.

The dependent variable is the standardized change in reported net income, and the union dummy takes a value of one in a union negotiation year and zero otherwise. Model 1 employs the proportionate change in earnings and cash flow relative to their 1979 values, each standardized by its 1979 value (1978 for U.S. Steel since it reported a loss in 1979). Model 2 employs the change in earnings and cash flow relative to their 1979 values, each standardized by 1979 stockholders' book equity. Model 3 employs the change in earnings and cash flow from the prior year, each standardized by the contemporaneous market value of equity. Model 4 employs the change in earnings and cash flow from the prior year, each standardized by the 1979 book value of equity. These regressions exclude two outlier years for LTV (while in bankruptcy) because they generate implausible (negative) values for the cash flow coefficient. Importantly, the sign and significance of the union dummy is *not* sensitive to exclusion of these outliers: with or without outliers, the dummy is significantly negative with p-values of 0.014 or better (the least significant t-value is -2.55). Models 1 and 2 employ 56 observations, and models 3 and 4 employ 55 (with the difference reflecting the fact that the LTV outliers affect three years of earnings changes calculated according to the latter two models).

Dependent variable:	$\frac{\text{Income}_t - \text{Income}_{79}}{\text{Income}_{79}}$	$\frac{\text{Income}_t - \text{Income}_{79}}{\text{Book equity}_{79}}$	$\frac{\text{Income}_t - \text{Income}_{t-1}}{\text{Market equity}_t}$	$\frac{\text{Income}_t - \text{Income}_{t-1}}{\text{Book equity}_{79}}$
Constant	−0.62	−0.06	0.05	0.05
	(−2.35)	(−1.74)	(0.90)	(1.51)
Cash flow variable	0.93	0.74	0.84	0.37
	(3.26)	(3.10)	(5.66)	(1.89)
Union dummy	−2.28	−0.24	−0.47	−0.31
	(−4.17)	(−3.29)	(−3.62)	(−4.71)
Adjusted R^2	36.0%	28.8%	56.4%	39.1%
F-statistic	16.47	12.14	35.91	18.37

billion – the five firms with negotiations reported an aggregate loss of $1.1 billion, and the remaining firm reported a profit of $0.4 billion. Two of six firms had union negotiations for which 1986 financial data are relevant, and they are responsible for $5.1 billion of the $5.8 billion aggregate loss reported by the six firms in that year.

Table 3 presents estimates of the following pooled time series, cross-sectional regression:

$$ChgNI = a + b_1 ChgCF + b_2 UNION,$$

where *ChgNI* is the size-standardized change in net income in a given year, *ChgCF* is the contemporaneous standardized change in cash flows, and *UNION* is a dummy variable that equals one if earnings are released during a union negotiation, and zero otherwise. As a robustness check, regression results are presented for four alternative specifications that differ in the

particular measures of (i) net income and cash flow changes and (ii) size standardization variables.

All specifications exhibit high overall F-statistics and a significant relation between the current period's cash flow and earnings changes. The positive cash flow coefficients do, however, depend on exclusion of two years of LTV data in which the firm had comparatively mild cash flow problems but (i) filed for bankruptcy (1986), and (ii) took huge pension-related writeoffs while attempting to shift its pension obligations to the federal government (1988).

Importantly, exclusion of the LTV outliers does *not* affect the sign or statistical significance of the union dummy – with or without outliers, union negotiations are associated with materially more negative declines in the reported earnings of sample firms. In all specifications in table 3, the union dummy is negative and strongly significant, with t-statistics ranging from -3.29 to -4.71, and the dummy exhibits comparable t-values when the LTV outliers are retained in the analysis. The union dummy also remains significantly negative under an error components analysis which adjusts for potential cross-sectional and time series dependence in the residuals. (Under the latter approach, t-statistics on the union dummy range between -2.42 and -3.14.)

Liberty and Zimmerman (1986) find no evidence of income-decreasing accounting choices by managers of firms in routine labor negotiations. Our finding of a significant relation is perhaps attributable to our focus on firms attempting to negotiate labor concessions, for which there is considerable other evidence of financial difficulty (for example, large-scale layoffs, managerial pay cuts, and dividend reductions). Relative to routine negotiations, requests for givebacks during visible signs of financial difficulty enable managers to depress income with fewer suspicions among rank-and-file workers that the claimed financial difficulties are simply the result of judicious accounting choices.

A closer look at unusual income items provides further evidence that managers systematically depressed reported earnings during union negotiations. For our sample, unusual items often include one-time special charges that reflect the real restructuring decisions made by sample managers during the 1980–1988 period. For example, a major capacity reduction achieved through plant closures requires that, *at some point*, substantial earnings charges be taken. Under current accounting rules, those charges can be taken in anticipation of, contemporaneous with, or after the economic event that triggers the earnings charge (the actual closing of plant doors), when managers *expect* the closing is permanent (a closed plant will not reopen).[14]

[14]See Strong and Meyer (1987), Elliott and Shaw (1988), 'Accountants Debate Tightening Rules for 'Big Bath' Writeoffs by Companies', *WSJ*, February 11, 1986, 'The Half a Hit Gambit', *Forbes*, December 31, 1984, 'Rumplestilzchen Accounting', *Forbes*, February 24, 1986, and 'Big Bath? Or a Little One?', *Forbes*, October 6, 1986. Concern about the scope of managerial discretion over the timing and magnitude of noncash writeoffs has recently led the Financial Accounting Standards Board to propose new accounting standards for asset impairments.

Even though the restructuring decision requires an *eventual* earnings charge, managers have incentives and opportunities to accelerate (or defer) the charge into negotiation periods, thereby lowering the firm's reported income and strengthening their bargaining position with unionized labor, both because of the announced earnings charge and the permanence of the associated layoffs.

To illustrate the degree of managerial discretion with respect to the timing of restructuring charges, consider those taken to 1982 earnings by Bethlehem Steel. Three days before the 1982 fiscal year end, the firm announced a special charge that ultimately totaled $1.05 billion, and that contributed to a 1982 net loss of $1.5 billion.[15] Of the total charge, $700 million was for the expected present value of termination benefits engendered by a permanent closing of facilities. Although the entire present value of these costs was charged to 1982 earnings, the primary cash flow impact will occur as benefits are paid in future periods. Another $160 million was taken for the writedown of productive facilities, effectively accelerating future years' depreciation charges into the current period. An additional $190 million was provided for 'other' expected costs (primarily for anticipated *future* losses before shutdown).

Bethlehem management blamed continuing losses at the affected plants on 'noncompetitive' wages and benefits and on competition from a 'continuing high level of imports'. The announcement of the proposed closings came on December 28, 1982, two months before a union agreement was reached (on March 2, 1983). The primary facility affected was the Lackawanna plant, which Bethlehem announced that it would close *by the end of 1983* as part of the restructuring plan. Given the distant date of the expected closing, and the late date in the 1982 fiscal year on which the restructuring was announced, it seems reasonable to believe that Bethlehem management accelerated announcement of the closings to take the $1.05 billion charge against 1982 earnings. Appendix B, which documents layoffs and plant closings, describes a number of cases in which the timing of these events differs from that of the related earnings charges.

Table 4 provides a yearly summary of aggregate unusual income items, gross of income taxes, juxtaposed against aggregate net income for each year from 1980 to 1988. We initially identified restructuring charges and other unusual income items (which can be negative or positive, for example, gains from sale of discontinued operations) by searching Compustat for special items (item 17) and for extraordinary items and discontinued operations (item 48). We confirmed all such items as steel-related by inspection of annual reports. For 1982, a union negotiation year for all sample firms, table

[15]This charge, incidentally, is not the largest in our sample. The largest is the $2.7 billion charge taken by LTV in 1986, the year it filed for bankruptcy. LTV also took earnings charges of over $1 billion in 1988, as did U.S. Steel in 1983 and 1986, and LTV made an accounting change with a −$2.3 billion impact on 1988 earnings.

Table 4

Aggregate unusual income items and reported net income for the seven major domestic steel producers, 1980–1988.

All unusual items are steel-related, are reported gross of income tax, and are taken from corporate annual reports for 1980–1988. Unusual items were initially identified from inspection of Compustat item 17, special items, and item 48, extraordinary items and discontinued operations. There are seven sample firms for 1980–1983 and six sample firms for the remaining years (because of the 1984 merger of LTV and Republic Steel).

Year	Number of firms with unusual items (number with) union negotiations)	Aggregate unusual items (in $ billions)	Aggregate net income for all sample firms (in $ billions)
1980	2 (0)	0.14	1.18
1981	3 (0)	0.09	2.30
1982	6 (7)	−1.95	−3.15
1983	6 (0)	−1.65[a]	−2.78[a]
1984	4 (0)	0.11	−0.32
1985	5 (5)	−0.68[b]	−0.65[b]
1986	5 (2)	−4.01[a,b]	−5.75[a,b]
1987	4 (0)	−0.06	1.11
1988	4 (0)	−3.67[a]	−1.61[a]

[a]The 1983 figures include a −$1.2 billion dollar charge taken by U.S. Steel for plant closures that the firm announced after the USW refused its request for concessions beyond those granted in the March 1983 contract agreement. The 1986 figures include a −$2.7 billion provision for discontinued operations taken by LTV the year it filed for bankruptcy. The 1988 figures include −$3.5 billion of charges taken by LTV, among them a −$2.3 billion accounting change for post-retirement benefits. As discussed in the text, LTV's 1986 and 1988 net income figures were treated as outliers for the regressions in table 3.

[b]In 1985, the five firms that held union negotiations reported aggregate income of −$1.06 billion (and unusual items of −$0.70 billion), while the remaining firm reported a profit of $0.41 billion (and had unusual items of +$0.02 billion). In 1986, the two firms that held union negotiations had total income of −$5.08 billion (unusual items of −$3.73 billion), while the remaining four firms had aggregate income of −$0.67 billion (unusual items of −$0.28 billion).

4 indicates that the initial aggregate industry loss of $3.15 billion includes $1.95 billion of steel-related unusual items that reduced reported income. Although sample firms would have shown an aggregate loss in 1982 without the special charges, these charges nonetheless constitute a material portion of the total reported loss.

For 1983, table 4 shows aggregate industry losses of $2.78 billion after steel-related unusual items of −$1.65 billion. Because multi-employer union negotiations concluded in March 1983, we do not classify 1983 as a union negotiation year. As detailed in appendix A, however, U.S. Steel in December 1983 requested additional concessions (beyond those in the March 1983 agreement) at several local plants. The USW refused on grounds that U.S. Steel was trying to generate unfair competition among local workers. Three days before the fiscal year-end, U.S. Steel responded by announcing five plant closures and taking a special charge to 1983 income of −$1.2 billion (73% of the 1983 aggregate unusual items in the sample). Thus, while our

regressions do not consider 1983 a union negotiation year, U.S. Steel's substantial earnings charges in that year are clearly related to its requests for additional labor givebacks at local plants.

In both 1986 and 1988, table 4 also shows very large unusual items relative to reported losses. The 1986 unusual items are due primarily to a large writeoff for discontinued steel operations (−$2.7 billion, or 67.3%, of aggregate unusual items) taken by LTV the year it filed for bankruptcy. When announcing its bankruptcy filing, LTV indicated it would seek labor concessions, and the writeoff was taken during negotiations over those concessions. The 1988 unusual items are almost entirely (98%) explained by an income-reducing accounting change (−$2.3 billion) and steel-related writeoffs (−$1.3 billion) taken by LTV as part of its attempt to shift its pension obligations to the federal government. (The 1988 actions were probably designed to help LTV's bargaining position, but with respect to the Pension Benefit Guaranty Corporation, rather than the USW.)[16]

5. Concessions by managerial and salaried personnel

We find pervasive evidence that sample firms' requests for union concessions are accompanied by sacrifices by top management and white-collar workers. Table 5 reports regressions of changes in CEO compensation (annual salary plus bonus) on changes in company performance (annual operating cash flows), and a dummy variable that equals one in union negotiation years and zero otherwise. As expected, greater cash flow is associated with higher CEO compensation for all models.

More importantly, the union dummy is uniformly negative and significant in all specifications, with t-statistics ranging from −4.55 to −4.93. We also find significantly negative values (with t-statistics from −2.48 to −3.56) for the union dummy under an error components analysis which adjusts for potential cross-sectional and time series dependence in the residuals. The significance of the union dummy is similarly robust to alternative specifications in which (i) CEO compensation is replaced by that for the firm's five highest-paid executives, and (ii) cash flow is replaced by alternative performance measures based on earnings or stock returns (details not reported here).[17] Thus, controlling for company performance, managerial compensa-

[16]LTV's figures for both 1986 and 1988 are treated as outliers in the regression results reported in table 3. As indicated earlier, inclusion of these two observations reverses the sign of the coefficient on cash flow changes, but does not alter the sign or significance of the union dummy variable.

[17]Given managers' tendency to depress earnings during union negotiations (table 3), we specify the current regressions with company performance measured by cash flows rather than earnings. The union dummy remains negative and highly significant when earnings changes (or stock returns) are used to measure company performance. The latter regressions show no significant relation between changes in executive compensation and changes in earnings (or stock returns).

Table 5

Parameter estimates (t-statistics) for pooled regressions of changes in chief executive officer salary and bonus on changes in cash flow and presence of union negotiations for the seven major domestic steel firms, 1980–1988.

In all specifications, the dependent variable is the proportionate change from the prior year in the CEO's annual salary plus bonus (per proxy statements), and the union dummy takes a value of one in a union negotiation year and zero otherwise. Models 1–4 differ according to the measure of standardized cash flow used as an explanatory variable. (Cash flow changes are taken from Compustat and company annual reports.) Model 1 employs the change in cash flow relative to its value in 1979. Model 2 employs the change in cash flow relative to its value in 1979, standardized by stockholders' book equity in 1979. Model 3 employs the change in cash flow from the prior year, standardized by the contemporaneous market value of equity. Model 4 employs the change in cash flow from the prior year, standardized by the book value of equity in 1979. All models employ 51 observations (three years of compensation change data for Republic and eight years for each of the remaining six firms). Although not reported here, the statistical significance of these results continues to hold when the dependent variable is defined as the proportionate change in salary plus bonus for the five highest-paid executives. For the latter regressions, the t-values on the union dummy range from -4.78 to -5.23.

Cash flow variable:	$\frac{CF_t - CF_{79}}{CF_{79}}$	$\frac{CF_t - CF_{79}}{\text{Book equity}_{79}}$	$\frac{CF_t - CF_{t-1}}{\text{Market equity}_t}$	$\frac{CF_t - CF_{t-1}}{\text{Book equity}_{79}}$
Constant	0.20	0.19	0.19	0.18
	(5.07)	(4.80)	(4.67)	(4.46)
Cash flow variable	0.08	0.37	0.10	0.34
	(2.69)	(2.27)	(1.94)	(2.05)
Union dummy	−0.37	−0.37	−0.37	−0.35
	(−4.93)	(−4.90)	(−4.74)	(−4.55)
Adjusted R^2	38.0%	35.5%	33.8%	34.4%
F-statistic	16.30	14.78	13.75	14.09

tion is significantly lower during union negotiations than for the same firms in nonnegotiation years.

This inference is strongly supported by the data in appendix C, which summarizes sacrifices by managerial and salaried personnel, as reported in the *WSJ*. The high incidence of such sacrifices is most evident for 1982, the year in which all seven sample firms simultaneously sought wage concessions from the USW. Appendix C indicates that, in that year, six of the seven major steel producers (all except LTV) made public announcements that were carried by the *WSJ* of managerial and white-collar pay cuts, wage freezes, and/or benefit reductions.

Overall, the *WSJ* articles (for 1982 and other years) indicate that large numbers of nonunion workers took pay cuts during union negotiations. Bethlehem Steel announced salary reductions for 250 high level managers (June 1982), compensation cuts for 22,000 salaried workers (July 1982), benefit reductions for all salaried employees (August 1982), and further pay cuts for 14,000 nonunion personnel (January 1983). U.S. Steel announced

wage cuts for 27,000 management and salaried personnel (June 1982), followed by further wage reductions seven months later. While announcing the latter, the firm's chairman stated that only the USW 'has refused to contribute to the sacrifices made by everyone else in the steel family to ensure the short- and long-term survival of the company...' (January 1983). All of these announcements were made during union negotiations (June 1982 to March 1983). (Announcements similar in both content and timing were made by Armco, Inland, Republic, and National Steel.[18])

A reading of appendix C suggests that, in percentage terms, financial sacrifices by top management commonly exceed – or at least are portrayed by the firm as exceeding – those required of the firm's other employees. For example, National Steel announced a pay freeze for 7,000 salaried personnel at the same time it announced pay cuts of 10% to 25% for senior management (March 1982). Republic Steel announced salary cuts of 7% to 10% for 5,800 employees while claiming a total compensation reduction of 18% for managerial personnel (August 1982). Similar emphasis on the sacrifices made by top management can be found in the appendix C entries for Armco (April 1982), Bethlehem Steel (June 1982), Inland Steel (April 1982), LTV (January 1986), and U.S. Steel (June 1982).[19]

Despite the pay reductions we document, managerial and white-collar compensation in some firms remained sufficiently 'generous' to draw public criticism from the steelworkers' union. For example, the USW attacked an interest-free loan to the CEO of LTV in the year of a record loss when, according to the union, the money could have been used to prevent the elimination of more than 600 factory jobs (May 1985). A union spokesperson reacted to Inland Steel's announcement that salaried workers' bonuses would be reduced by indicating that the payment of any bonuses at all would 'make us take a good hard look at the company's contract proposals next year.... If they have such a severe cash flow problem, how can they afford to give bonuses to salaried personnel?' (December 1982). And Bethlehem Steel's CEO was apparently forced to retire early in part because his salary increases engendered strong resentment among both blue- and white-collar workers, who had themselves suffered multiple wage cuts (May 1986).

[18]While we found no *WSJ* report of sacrifices by nonunion personnel at LTV during 1982, we did find evidence of wage and benefit reductions for 8,000 LTV salaried employees in August 1985, before union negotiations opened that December. LTV later announced a further round of wage reductions, as well as headquarters staff layoffs (January 1986).

[19]Bethlehem Steel provides a clear illustration of how top management used its own financial sacrifices to argue for similar sacrifices from other workers. When the firm's chairman announced a 10% salary reduction for upper management and a 5% cut for salaried workers, he also indicated that similar contributions by hourly employees were needed to return the firm to profitability and to provide employment security for as many people as possible (September 1985).

Table 6

Salary plus bonus payments to the chief executive officer and the five highest-paid executives of the seven major domestic steel producers, 1980–1988.

Executive compensation data are taken from proxy statements or forms 10-K.

	Chief executive officer	Top five executives
(A) *Mean (median) percentage change in salary plus bonus*		
Nonnegotiation years	19.5% (16.2%)	16.0% (9.0%)
Union negotiation years	−18.1% (−19.2%)	−15.4% (−15.7%)

(B) *Annual breakdown of salary plus bonus*

Year	No. of firms with union negotiations	Mean (median) salary plus bonus	
1980	0	$518,000 ($479,000)	$1,691,000 ($1,492,000)
1981	0	708,000 (664,000)	2,213,000 (1,986,000)
1982	7	489,000 (425,000)	1,591,000 (1,437,000)
1983	0	477,000 (419,000)	1,502,000 (1,388,000)
1984	0	603,000 (569,000)	2,008,000 (1,867,000)
1985[a]	5	609,000 (546,000)	1,945,000 (1,825,000)
1986[a]	2	552,000 (506,000)	1,813,000 (1,657,000)
1987	0	702,000 (585,000)	2,150,000 (1,832,000)
1988	0	908,000 (753,000)	2,774,000 (2,489,000)

[a]For the five firms for which 1985 data are relevant for union negotiations, the average change in CEO compensation is 2.4% (median, −0.8%), while the remaining firm had an increase of 3.8%. For the two firms for which 1986 data are relevant for union negotiations, the mean and median change in CEO compensation is −15.3%, while it is −0.4% (median, 2.4%) for the remaining firms.

Although these reports indicate continued worker resentment of managerial pay, the general pattern in appendix C is one of significant compensation sacrifices by top management during union negotiations. Table 6 reports quantitative data that support this conclusion. Panel A documents that, over the period 1980–1988, annual salary plus bonus payments for CEOs increased an average of 19.5% in years without union negotiations (median increase, 16.2%). In strong contrast, average CEO salary plus bonus decreased 18.1% in union negotiation years (median decrease, 19.2%). The same pattern holds for salary plus bonus payments to the five highest-paid officers: on average, they increased 16.0% in nonnegotiation years and decreased 15.4% in union negotiation years (with medians of +9.0% and −15.7%, respectively).

The relation between union negotiations and compensation sacrifices by top management is also manifested in the year-by-year total cash compensation reported in panel B of table 6. Most strikingly, average CEO salary plus bonus fell from $708,000 (median $664,000) in 1981, when union negotiations were not held, to $489,000 (median $425,000) in 1982, when all firms held

Table 7

Bonus payments to chief executive officers of the seven major domestic steel producers, 1980–1988.

Bonus payments to CEOs are taken from company proxy statements or 10-K forms. Entries marked 'n.d.' indicate that bonuses were not separately disclosed in these sources. Data for Republic are reported through 1983, after which it was merged with LTV. The figures for LTV are for the firm's Corporate Performance Reward Plan, which the company labels its 'bonus' plan. LTV also has a Management Variable Compensation Plan, which the firm claims is not a bonus plan (a claim that was questioned when it generated cash payments to executives while the firm was in bankruptcy). The latter plan provides 25–50% of an individual's base salary if unspecified personal and corporate goals are met. Under that plan, payments received by LTV's CEO were $165,625 in 1985, $130,000 in 1986, $130,000 in 1987, $265,000 in 1988, and were not disclosed in 1980–1984. An asterisk indicates the firm was involved in union negotiations in the specified year.

	U.S. Steel	LTV	Bethlehem	Armco	National	Republic	Inland
1980	$180,000	n.d.	n.d.	n.d.	n.d.	$50,000	$165,000
1981	290,000	n.d.	$175,000	n.d.	$217,876	200,000	200,000
1982	0*	n.d.*	0*	0*	0*	0*	n.d.*
1983	0	n.d.	0	0	0	0	n.d.
1984	300,000	n.d.	0	0	200,000	—	n.d.
1985	275,000	0*	0*	0*	0*	—	0*
1986	n.d.*	0*	0	0	0	—	n.d.
1987	600,000	0	0	0	0	—	n.d.
1988	1,000,000	0	106,000	48,400	0	—	n.d.

negotiations. In 1982, average CEO salary plus bonus decreased by 30.9% (the median decreased by 36.0%).[20] The five highest-paid executives had salary plus bonus reduced from an average $2,213,000 (median $1,986,000) in 1981 to an average $1,591,000 (median $1,437,000) in 1982. In 1982, the average salary plus bonus for the five highest-paid officers decreased by 28.1% (the median decreased by 27.6%).

An interesting question is whether managerial pay cuts are driven by the loss of accounting-based bonuses in union negotiation years with especially low reported income. Table 7 reports bonus payments to individual firms' CEOs for the 1980–1988 period. As with firms in general, proxy statement bonus disclosures are sporadic for our sample. They nonetheless reveal the loss of material bonuses for CEOs of four firms (U.S. Steel, Bethlehem, National, and Republic) in the union negotiation year of 1982. Strikingly, for every firm in every negotiation year (marked by an asterisk in the table), the

[20]For firms for which 1985 and 1986 data are relevant for union negotiations, the average changes in CEO compensation are, respectively, 2.4% and −15.3% (medians, −0.8% and −15.3%). For firms that did not hold negotiations, the corresponding years' average changes are 3.8% and −0.4% (medians, 3.8% and 2.4%).

CEO either received no bonus or the bonus was not separately disclosed.[21] This fact suggests that proxy disclosures are designed to reveal personal costs incurred by the CEO (by reporting that no bonus payments were made), or to diminish the likelihood of adverse publicity over CEO compensation (by not reporting bonus amounts if paid).

We find little evidence that the managerial compensation decline reported in table 6 was caused by accounting-based bonus formulas (coupled with lower reported earnings).[22] While incomplete proxy disclosures make it difficult to draw strong conclusions on this issue, we did find three cases in which the bonus lost by a CEO in a union negotiation year was larger in absolute value than the decline in salary plus bonus. In one of these cases, however, the firm had a positive bonus pool which the board of directors chose not to distribute currently. In the remaining two cases, proxy disclosures suggest that the firm's bonus plan gave the board the discretion to specify a positive bonus pool, but the board chose not to pay bonuses. Thus, even in these cases, discretionary board actions rather than explicit bonus plan formulas are apparently responsible for the observed decline in total salary plus bonus.

Finally, although compensation reductions were clearly substantial for (the relatively few) top managers, their dollar magnitude is obviously small relative to the total dollar losses of the many thousands of union and white-collar workers who lost their jobs or experienced wage and benefit reductions. Clearly, these executive compensation reductions will not directly generate material improvements in corporate cash flow. Their indirect effects may be substantial, however, since they apparently serve to show workers that individual executives believe sufficiently in their plans to deal with the firm's financial troubles that they too are willing to bear personal financial sacrifices.[23] In this view, managerial pay cuts are not important for the dollars they save per se, but rather because they represent symbolic sacrifices that encourage all stakeholders to participate in the concessions needed to salvage the firm.

[21]For example, U.S. Steel's proxy statements report that the CEO's bonus fell from a substantial $290,000 in 1981 to zero in the union negotiation year of 1982. The firm's proxy statements do not disclose the CEO's bonus for 1986 (also a negotiation year), even though bonuses were disclosed in all other years and large bonuses were paid in the two prior and two subsequent years.

[22]Our earlier analysis suggests that the firm's reported income is at least partly under management control, so that bonus-plan-induced cuts in managerial pay are not appropriately viewed as exogenous to managers. Rather, our findings suggest that special earnings charges and foregone bonuses are elements of a strategy chosen by managers to portray the firm as in serious trouble.

[23]It is possible that managerial pay was cut for other reasons as well. For example, the board of directors may have judged business conditions as sufficiently bad to require sacrifices at all levels of the organization. Another (not mutually exclusive) possibility is that executive pay cuts reflect the board's assessment that managerial performance was especially poor.

6. Concessions by shareholders

Dividend reductions serve two complementary functions in negotiating labor concessions. Because managers are reluctant to cut dividends and risk antagonizing shareholders, a dividend reduction helps management make a credible case to the union that the firm is indeed experiencing serious financial problems. Second, members of the union rank-and-file are likely to interpret dividend cuts as concessions by shareholders. Finance theory, of course, treats the total stock market return (dividend plus equity value change) as the complete measure of shareholders' compensation in a given period. However, the dividend component of shareholders' total return is more likely to be viewed by the rank-and-file as shareholder 'wage' payments analogous to cash wages that workers receive from the firm.[24]

Table 8 reports the timing of dividend reductions announced during the 1980–1988 period by sample firms and the two-day stock returns associated with each announcement. The table indicates that all seven firms reduced dividends during their financial difficulties, and that all except U.S. Steel eventually omitted dividends altogether. (Although Republic Steel did not omit dividends as an independent firm, LTV did so shortly after it acquired Republic.) The high incidence of dividend omissions is notable because (i) the median firm had paid dividends for 46 continuous years,[25] and (ii) managers of firms with long dividend histories appear especially reluctant to break a string of continuous dividend payments [DeAngelo and DeAngelo (1990)].

Ten of the 21 dividend reductions reported in table 8 occurred in 1982, the year in which union negotiations were opened early (in June) to accommodate steel firms' request for labor concessions.[26] Two firms (Bethlehem Steel and Armco) announced additional dividend reductions in January 1983, just before the USW agreed to wage concessions in March. In all, 12 of the 21 dividend reduction announcements reported in table 8 came in the period immediately preceding the USW's initial concessionary wage agreement. Moreover, all seven sample firms announced dividend reductions during these particular union negotiations. These dividend reductions ranged from a cut of 50% (by U.S. Steel, LTV, Republic, and Inland Steel) to a cut of 87.5% (by National Steel) during the negotiation period (June 1982 to March 1983).

[24]In contrast, equity value changes represent the judgments of stock market traders and involve no direct resource distributions. Our focus on dividend reductions is conservative insofar as shareholders also experienced significant equity value declines (see below).

[25]Sample firms had paid dividends for a range of two (for LTV) to 58 (for National Steel) continuous years at the time they omitted dividends.

[26]Of these ten dividend reductions, eight were announced after the negotiations opened in June. The other two (by Republic and Inland Steel) came about one month earlier.

Table 8

Timing of dividend reductions and two-day stock returns at dividend reduction announcements by the seven major integrated steel producers, 1980–1988.

Stock returns are calculated over the day of and day after each dividend announcement. The returns are taken from the CRSP daily tape and are not adjusted for market movements. An asterisk indicates the dividend reduction announcement came during union negotiations.

Date	Dividend reduction	Two-day stock return
	U.S. Steel Corporation	
10/27/82*	Cuts $0.50 dividend to $0.25	−7.88%
	LTV Corporation	
11/15/82*	Cuts $0.125 dividend to $0.0625	−12.13
10/29/84	Omits dividend (after 2 years)	−12.22
	Bethlehem Steel Corporation	
7/29/82*	Cuts $0.40 dividend to $0.25	−0.79
1/27/83*	Cuts $0.25 dividend to $0.15	−3.51
1/31/85	Cuts $0.15 dividend to $0.10	−6.78
10/31/85	Omits dividend (after 46 years)	−18.06
	Armco Inc.	
7/26/82*	Cuts $0.45 dividend to $0.30	−0.78
1/31/83*	Cuts $0.30 dividend to $0.20	−1.47
4/25/83	Cuts $0.20 dividend to $0.10	−9.65
7/30/84	Omits dividend (after 44 years)	−13.67
	National Steel Corporation	
8/21/80	Cuts $0.675 dividend to $0.50	−4.66
6/17/82*	Cuts $0.50 dividend to $0.25	−4.00
12/16/82*	Cuts $0.25 dividend to $0.0625	−10.03
6/23/88	Omits dividend (after 58 years)	−2.16
	Republic Steel Corporation	
5/13/82	Cuts $0.50 dividend to $0.25	−5.66
11/17/82*	Cuts $0.25 dividend to $0.125	+2.31
	Inland Steel Company	
7/30/80	Cuts $0.70 dividend to $0.50	0.00
4/29/82	Cuts $0.50 dividend to $0.25	−8.02
10/28/82*	Cuts $0.25 dividend to $0.125	−3.01
9/27/85	Omits dividend (after 51 years)	−5.91

As expected, sample firms' dividend reduction announcements are associated with negative stock returns. An interesting new finding is that the average stock return of −4.13% for dividend reduction announcements during union negotiations is materially less negative than the average −7.89% return for reductions announced in nonnegotiation periods. These mean returns differ significantly at the 0.048 level under a one-tailed t-test. The median returns of −3.26% and −6.78% (for negotiation and nonnegotiation

periods, respectively) differ significantly at the 0.049 level under a one-tailed Wilcoxon rank sum test.

The difference in average share price reactions may reflect the fact that, because steel firms' troubles were well known in negotiation years, stock market traders could better predict dividend reductions, hence capitalize them prior to public announcement. Another possible explanation is that dividend omissions (which for our sample came during nonnegotiation periods) have more negative share price consequences than do substantial cuts to a still-positive level.[27] Yet another possibility is that traders interpret dividend cuts during union negotiations as entailing less 'bad news' because in part they represent negotiating tactics designed to obtain givebacks from labor; the positive cash flows expected from these concessions partially offset the bad news usually signaled by a dividend cut.

Analysis of *WSJ* reports indicates that managements' announcements of dividend reductions sometimes occur simultaneously with, or are linked explicitly to, sacrifices by unionized labor, white-collar workers, and top executives. More commonly, however, management explains the decision to reduce dividends as a cash conservation measure that was prudent and necessary in light of the firm's financial difficulties. This fact probably reflects the reality that management is ultimately the shareholders' agent, and must therefore be careful not to anger that constituency by appearing to retain cash that will in effect subsidize unionized labor.[28] It may also help explain why managers' case for union concessions is typically based on the firm's general financial difficulties (which is buttressed by their willingness to risk further shareholder dissatisfaction by cutting the dividend).

Although dividends were reduced substantially in percentage terms, the cash conserved through dividend reductions pales in comparison with the labor cost savings achieved through layoffs and wage cuts. The seven firms paid $463 million in common stock dividends in 1981 which, after numerous dividend reductions in 1982, declined to $201 million in 1983. Although certainly not negligible, this $262 million dividend decline is modest in relation to sample firms' reduction in aggregate annual labor costs, which total in the billions of dollars.[29] These facts indicate that cash conserved via dividend reductions, like that from managerial pay cuts, was at best a

[27]This argument may help explain the difference in mean stock returns, but less plausibly explains the significant difference in sample medians.

[28]Some evidence supporting this interpretation can be found in the appendix C entry for Armco (April 1983).

[29]This conclusion is reinforced by the fact that labor cost savings are annual amounts that tend to persist, while cash savings from dividend reductions are more likely temporary. Bethlehem, Armco, and Inland reinstated dividends in 1988 or 1989, albeit at amounts lower than paid before the initial cuts, while USX increased its dividend to 70% of what it had been. Only National and LTV continue to pay zero dividends at this writing.

second-order consideration. Rather, the importance of these reductions seems to lie in their *indirect* cash consequences – they foster a general cooperative spirit that all stakeholders are making material sacrifices, which encourages organized labor to make (in the aggregate, much larger) concessions of their own.

7. Summary and conclusions

This paper studies the effect of union negotiations on the corporate policies of the seven major domestic steel producers during the 1980s, a period in which strong competitive pressures to reduce capacity led managers to negotiate major concessions from unionized labor. The resultant restructuring of these firms generated strikingly large reductions in productive capacity, wage bills, and total employment. Over the period 1980–1988, the seven sample firms reduced productive capacity by 38.2%. Their annual aggregate wage bill declined by 46.6% from about $16.1 billion in 1980 to around $8.6 billion in 1988, a decline achieved largely through radical employment reductions. In total, sample firms reduced employment by almost two-thirds of their work force, for a loss of about 300,000 jobs over the decade.

Managers of domestic steel firms used a variety of approaches to help convince organized labor that their companies were in serious trouble, and therefore required major concessions. From the perspective of rank-and-file workers, the most convincing indication of serious trouble was probably the large-scale layoffs we document. We also find evidence that managers used reported losses and financial sacrifices by other (nonunion) stakeholders to support their case for labor concessions. Net income is materially lower during union negotiations than in other periods, when we control for cash flow performance. And although sample firms in the aggregate had positive cash flows in every year from 1980 to 1988, they reported losses in each of the years 1982–1986 and again in 1988. Importantly, the losses reported in union negotiation years are to a large degree driven by special earnings charges that reflect managers' real restructuring decisions, but whose timing is discretionary across negotiation and nonnegotiation years.

Controlling for company performance, salary plus bonus payments to CEOs (and to the five highest-paid executives) are significantly lower during union negotiations than for the same firms in other years. CEOs experienced an average 18.1% decline in salary plus bonus in union negotiation years, which contrasts markedly with their average increase of 19.5% in nonnegotiation years. For the five highest-paid executives, salary plus bonus declined an average 15.4% in union negotiation years, which compares with an average 16.0% increase in other years. Although managerial pay cuts are large in percentage terms (relative to the pay reductions of union and nonunion workers), their absolute magnitude is in the hundreds of thousands of dollars. The pervasive reductions in cash dividends we observe aggregate to the hundreds of millions. Both pale when compared with the billions of dollars in annual labor cost savings achieved through layoffs and wage reductions.

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Reductions in the compensation paid top management – although small in relation to sample firms' total labor cost savings – apparently serve as symbolic actions that help foster a general attitude that financial sacrifices are required by all stakeholders. Consistent with this view, white-collar pay cuts and reductions in cash dividends are substantial on a percentage basis, pervasive, and clustered during union negotiations. Analysis of top management's public statements indicates they cited low (and diminished) earnings, dividend cuts and omissions, white-collar pay cuts and layoffs, and reductions in their own compensation to portray the firm as in serious financial difficulty, hence in need of substantial concessions from unionized labor. Overall, then, our evidence suggests that the corporate policies of domestic steel firms requesting labor concessions during the 1980s were determined by strategic contracting considerations as well as by the firm's technological capabilities, product market pressures, etc.

Appendix A

Chronology of union negotiations for the U.S. steel industry, 1980–1988

This chronology was developed from information in *Standard & Poor's Industry Surveys* and the *Wall Street Journal*. Each entry contains a brief description of the relevant union activity during a given month, sometimes taken verbatim from the *WSJ* article.

February 1980: The United Steel Workers (USW) open contract negotiations with the steel industry.

April 1980: Contract negotiations are settled, with wage and benefit hikes that are estimated to increase labor costs by 37% over the next three years.

July 1981: Wheeling-Pittsburgh Steel is forced out of the Basic Industry Bargaining Group after negotiating a separate labor contract with the USW.

June 1982: USW local presidents authorize bargainers to begin informal talks with the industry even though the current labor contract is not set to expire until August 1, 1983.

July 1982: Union negotiations break down.

November 1982: Labor and management reach a tentative agreement, but their proposal fails to win the approval of local union presidents, raising the possibility of an industry-wide strike.

December 1982: Wheeling-Pittsburgh Steel negotiates a 43-month labor contract at wage reductions of $3.65 per hour (over and above the $1.60 per hour reduction the firm had negotiated in 1981) in exchange for a promise to invest the labor cost savings in steel operations, 50% sharing of pretax profits after debt service charges and dividend payments, and the issuance of preferred stock to an employee trust fund.

February 1983: Union negotiations reopen.
March 1983: The seven major steel firms reach agreement with the USW. The agreement reduces labor costs by $1.25 per hour in wages, eliminates certain benefits, and provides for changes in the work rules. Concessions are to be restored during the 41-month life of the contract.
December 1983: U.S. Steel requests further contract concessions at five plants it is considering closing, even though multi-employer negotiations concluded earlier in the year. The USW denies the request, saying that U.S. Steel is unfairly attempting to engender competition among local workers. In response, U.S. Steel closes a number of plants and reduces other operations for a total capacity reduction of 20%, taking a $1.2 billion writeoff and reducing employment by 15,400. Of the five plants at which concessions were denied, three were closed and another had its capacity reduced substantially.

August 1984: The steelmakers' bargaining group begins to collapse, following National Steel's announcement that it will bargain alone when the current contract expires in two years.

April 1985: Wheeling-Pittsburgh Steel files for bankruptcy.
May 1985: The steel industry bargaining group collapses after a 30-year history. LTV announces its intent to seek wage and benefit concessions from the USW.
September 1985: LTV suspends dividends on four issues of preferred stock.
October 1985: Bethlehem omits its dividend for the first time in 46 years and says it will seek to restructure its agreement with the USW 'well before' the current contract expires next July 31.

January 1986: LTV and Bethlehem begin separate negotiations with the USW on the contract set to expire August 1, 1986.
February 1986: Armco, National, and Inland set late March target dates to negotiate new labor contracts. The USW defers scheduled wage increases for LTV, but not for Bethlehem and National.
March 1986: LTV reaches a tentative agreement with the USW. Armco, National, and Bethlehem, discouraged over the terms of LTV's labor pact and its uncertain ratification, abandon talks. Their concern is that, by negotiating first with the weakest steel producer, the USW has gained concessions that it can then impose on other steelmakers. Inland will keep bargaining.
April 1986: Inland and the USW break off talks after failing to reach an agreement by the target deadline. The USW approves the contract with LTV, which provides for a $3.15 per hour reduction in labor costs with a payback schedule over time.
May 1986: Bethlehem reaches a tentative accord with the union that will reduce labor costs by $1.97 per hour for one year.
June 1986: U.S. Steel, the last of the major steel firms to negotiate with the USW, opens bargaining talks. The USW approves the Bethlehem agreement. Inland Steel signs a tentative three-year pact that freezes wages, cuts benefits, and gives workers profit sharing. Armco extends the August 1 contract expiration date for three months.
July 1986: LTV files for bankruptcy protection, and seeks additional concessions from the USW. The USW strikes one of the firm's plants for a week to protest cancellation of retirees' health and life insurance benefits until LTV reinstates them for six months. Bethlehem omits the dividend on two issues of preferred stock, citing the LTV bankruptcy filing. The Inland agreement receives final approval. U.S. Steel negotiations break down.
August 1986: The USW strikes U.S. Steel.
September 1986: Armco threatens to file for bankruptcy as local steelworkers strike the firm's main plant in Middletown, Ohio. The company backs off its demands for wage concessions to settle the strike after five days with a new contract that provides only a wage and benefit freeze. Bankrupt LTV seeks to renegotiate its most recent pact with the union.
October 1986: U.S. Steel resumes bargaining with the USW, as does Armco.
November 1986: Armco reaches agreement on a new labor contract that will vary from mill to mill. U.S. Steel breaks off contract talks.

January 1987: U.S. Steel reopens union negotiations, and the strike ends.
February 1987: The union ratifies a four-year contract with U.S. Steel that provides for wage reductions and profit sharing.
April 1987: The USW agrees to negotiate a new labor agreement with LTV by May 1, terms to be retroactive to February 1.
May 1987: The USW rejects a new pact with LTV, citing provisions that would eliminate jobs and management's failure to sacrifice as much as steelworkers have.
June 1987: LTV and the USW resume talks, and reach tentative agreement. The Pension Benefit Guaranty Corporation threatens to return LTV's pension obligations to the company because the labor agreement restores certain supplemental pension benefits to the steelworkers, while the government continues to bear the bulk of LTV's pension obligations.
August 1987: The USW ratifies the new contract with LTV despite the PBGC's continued legal challenge.

Appendix B

Employment changes and plant closures announced by the seven major domestic steel producers, 1980–1988

This appendix documents the net change in total employment (second column) during each of the years 1980–1988, as disclosed in the annual reports of the seven major domestic steel producers. If total employment figures are not disclosed, n.d. appears following the relevant year. Changes in total employment during union negotiation years are marked with an asterisk. The text provides details of those changes, particularly plant closings, as reported in corporate annual reports, the *Wall Street Journal Index*, and referenced articles.

Armco Inc.

1980:	+5,153	The company operates eight major steel plants at Ashland, Kentucky; Houston, Texas; Ishpeming, Michigan; Kansas City, Missouri; Marion, Ohio; Middletown, Ohio; Sand Springs, Oklahoma; and Mexico City, Mexico. In 1980, it lays off 770 workers at its main Middletown plant.
1981:	+7,685	The firm sells the Sand Springs, Oklahoma steel works and related fabricating plants, and ceases operations at the Marion, Ohio plant, taking a $3 million special charge to income. About 300 jobs will be lost if the firm cannot find a buyer for the Marion plant.
1982:	−15,647*	The firm offers early retirement, which 1,000 employees accept, for a charge to income of $31 million. In January 1983, the company announces its intent to cease operations at portions of its Houston, Kansas City, and Middletown plants, for a charge to 1982 income of $188 million. About 2,200 jobs are expected to be eliminated. The layoffs at the Kansas City plant are related to a refusal by a USW local to accept wage concessions. The layoffs at the Middletown plant total 1,070, and an additional 150 are cut at corporate headquarters. Of the total of 15,647 employee reductions in 1982 about 3,000 are permanent, according to the CEO.

1983:	−3,942	Steel-related special charges to 1983 income are $268.0 million for amounts anticipated to be incurred for the closing of the Houston steel mill in January 1984, for the anticipated combination of some operations of the Middletown and Ashland plants, and for the expected phaseout of the open hearth furnaces at Middletown and the hot strip mill at Ashland. The Houston plant closing is expected to idle 1,100 workers (the plant had employed 4,500 workers in 1981).
1984:	−3,688	The Houston plant is shut down and open hearth furnaces are closed at the Middletown plant, for a 30% reduction in total steel productive capacity during 1984. Headquarters staff is cut by an additional 170 persons.
1985:	−13,583*	The firm announces the closing of its Ambridge, Pennsylvania plant in the third quarter, taking a $110 million special charge and eliminating 1,200 jobs. The plant is closed in December 1985. The firm eliminates an additional 250 jobs at corporate headquarters.
1986:	−9,900	The firm takes a $235.0 million writedown on a joint venture with bankrupt LTV, and an additional $11.8 million restructuring charge on steel operations.
1987:	−300	In December, the firm announces the shutdown of certain unprofitable operations at the Kansas City plant. This action and the restructuring of other steel facilities result in a $79.2 million special charge, and reduce employment by 950.
1988:	−1,100	The firm agrees to sell its Houston plant, which was closed in 1983, to Northwestern Steel & Wire Company. Steelmaking capacity is about 5.8 million tons, down from 9.7 million tons in 1980. Armco now employs 19,500 workers, down from 59,975 in 1980.

Bethlehem Steel Corporation

1980:	−8,500	The firm operates eight raw steel producing plants at Bethlehem, Johnstown, and Steelton, Pennsylvania; Sparrows Point, Maryland; Lackawanna, New York; Burns Harbor, Indiana; Seattle, Washington; and Los Angeles, California. The firm announces a number of temporary shutdowns during 1980.
1981:	−5,400	The salaried workforce is reduced by 5%. The company announces a number of temporary closings and layoffs, including the shutdown of the Lackawanna plant for a week, which idles 600 workers.
1982:	−16,900*	The salaried work force is reduced by another 15%. The firm announces plans to close its Los Angeles plant and its intention to sell it and the Seattle plant, which it will operate until sale. It also announces plans to discontinue basic steelmaking at Lackawanna by the end of 1983 and to consolidate other operations. When these plans are fully implemented, the firm will have reduced total steelmaking capacity by 20% and eliminated 10,000 jobs. The charge to 1982 earnings is $1.05 billion for anticipated costs and losses (of which $700 million is the present value of employment costs, e.g., termination benefits, pensions, etc.). The company currently has 30,000 of 82,000 employees on layoff.
1983:	−14,100	The salaried work force is reduced by an additional 18%, for a total of almost 40% since the beginning of 1981. Basic steelmaking is discontinued at Lackawana in October, eliminating 7,300 jobs, after the firm rejects a USW offer to make wage concessions to keep the plant open.

1984:	−1,400	The salaried work force is reduced by 12%. The company sells the Seattle plant on December 31.
1985:	−6,900*	The salaried work force is reduced by 20%. The firm is attempting to narrow losses in its troubled bar, rod, and wire division by laying off 45% of the salaried work force and 35% of the hourly workers. It will close its Lebanon, Pennsylvania specialty steel plant, idling 600 people. Earnings charges for discontinued operations related to steel production are $87.4 million.
1986:	−7,000	The firm continues to lay off workers. It provides $28.0 million for discontinued steel operations, reducing its capacity from 17.6 to 16.0 million tons.
1987:	−3,100	No layoffs or shutdowns are announced.
1988:	−1,500	The company provides $113.0 million for discontinued steel and steel-related operations. Steelmaking capacity is 16 million tons, down from 21.5 million tons in 1980. It now employs 32,900 workers, down from 89,200 in 1980.

Inland Steel Company

1980:	−2,438	Inland has one major steel producing facility, the Indiana Harbor Works in East Chicago, Indiana, with 23,250 employees. The plant's 1980 steelworker layoffs peak at 2,900 in July.
1981:	+191	The Melrose Park, Illinois structural fabricating plant is closed in December, with no material earnings impact. The plant has 75 salaried and 300 hourly workers; the former had been on strike as members of a USW local.
1982:	−4,089*	The firm announces a number of temporary layoffs during the year. About 10% of the total workforce is on layoff, with 5,200 of Inland's 18,000 union workers idle. About 450 salaried personnel take advantage of an early retirement program, for a $9.9 million earnings charge.
1983:	−2,305	The company decides to close three raw materials facilities. The Jackson City Iron Ore Company mine (shut down in April 1982) will be permanently closed at an earnings charge of $15.6 million. The Lancashire No. 25 coal mine will be shut down, for a loss of 279 jobs and an earnings charge of $36.8 million. The firm is currently in negotiations to sell the Bishop Coal Company, and is taking a $6.0 million earnings charge for idle costs expected to be incurred prior to sale.
1984:	−1,779	The firm announces that it expects further across-the-board layoffs (about 24% of its 15,166 hourly employees are currently idle). It takes a 1984 earnings charge of $28.0 million for the planned reduction of 700 salaried jobs (14%) during 1985. It anticipates the closing of older steel facilities over the next three years, for a 1984 earnings charge of $27.0 million. The firm provides an additional $10.0 million to close part of its 'C' Coke Battery complex (shut down since October 1983) and $7.0 million for idle costs and losses on sale of its Bishop Coal Company (sold on December 31, 1984).
1985:	−2,508*	The firm announces it will shut down 30% of its capacity and lay off another 900 hourly workers. It takes a $32.0 million additional charge for the closure of its 'C' Coke Battery complex. It also provides $18.0 million for reductions in its salaried work force for another 300 employ-

		ees, and $33.3 million for permanent closure of a raw materials facility. It announces its intention to divest the Construction Products business segment and recognizes anticipated losses of $31.0 million.
1986:	−1,745	The company takes a $9.5 million earnings charge for the expected disposition of raw materials interests and $15.5 million of losses and expected losses on disposition of its Construction Products segment.
1987:	−1,928	The company takes a $23.0 million earnings charge for the planned shutdown over the next three years of older steel facilities.
1988:	−101	The obsolete 10-inch bar mill is closed. The company's steelmaking capacity is 6.5 million tons, down from 8.6 million tons in 1980. It now employs 20,639 workers, down from 34,903 in 1980.

LTV Corporation

1980:	−6,380	The company's steel operations are carried out primarily through its Jones & Laughlin Steel subsidiary, which has major plants at Pittsburgh and Aliquippa, Pennsylvania; Cleveland and Canton/Massillon, Ohio; the Indiana Harbor Works at East Chicago, Indiana; Chicago and Hennepin, Illinois; and Warren, Michigan. In 1980, it closes indefinitely blast furnaces at Cleveland, Aliquippa, and Indiana Harbor, and plans to close permanently a portion of its Pittsburgh plant.
1981:	−10,600	The firm closes temporarily a blast furnace at Indiana Harbor, and also plans to close temporarily some bar-producing facilities in Ohio.
1982:	−22,400*	The firm announces a number of temporary layoffs and the permanent closing of a coke battery near Youngstown, Ohio, taking a special charge of $30.0 million for consolidation of steel facilities.
1983:	+6,700	The firm plans to close permanently its stainless steel operations in Michigan.
1984:	+21,300	The company merges with Republic Steel. It announces numerous layoffs at a number of plants, and is said to be firing and forcing early retirement on managerial and salaried personnel after the merger. As a condition for merger approval, the Justice Department forces the company to divest the Massillon (owned by LTV) and Gadsden (owned by Republic) plants.
1985:	+1,800*	The company idles primary operations at the Aliquippa steel mill, and takes a $380.0 million special charge to 1985 earnings. A number of smaller, less efficient bar plants are also closed, and the last of the company's continuous-weld pipe operations are idled. Steelmaking capacity is reduced by one million tons by indefinitely idling some operations at the Chicago Works. The firm agrees to sell the Gadsden plant to its 2,200 employees, but this action is overruled by the Justice Department, which favors a buyer that plans to lay off all employees and will not bargain with the union.
1986:	−8,500*	The company files for Chapter 11 bankruptcy, and cancels health benefits and life insurance for its 76,000 (61,000 steel) retirees. In response, 4,400 USW workers strike the Indiana Harbor Works, LTV's most profitable plant, which supplies one-fifth of the firm's steel output. After a week, LTV agrees to provide alternative coverage. During the year, the firm idles four bar and pipe mills and other unprofitable facilities, and reduces its steelmaking capacity by 31%. It also rejects high-cost executory contracts and terminates its pension plans. It takes a

		$2.7 billion charge to 1986 earnings, which includes estimated claims associated with the pension termination, writedowns of productive assets, and anticipated costs and losses from the bankruptcy filing and related actions.
1987:	−100	The Pension Benefit Guaranty Corporation reinstates the firm's pension plans, forcing $2 billion in pension liabilities on LTV. It does so because, it argues, the firm's recent union agreement sweetens pension plans the government is subsidizing. LTV contests the action. The company announces it is seeking a buyer for its Warren, Ohio integrated steel plant.
1988:	−9,400	The company sells the Warren plant and announces its intent to sell its steel bar division, which employs 5,400 people. Its earnings charges include $950 million for Chapter 11 claims against steel operations, $350 million for the discontinuance of the bar division and other steel rationalization costs, and the $2.3 billion effect of a voluntary accounting change to recognize its liability for steel-related postretirement health benefits. The firm's steelmaking capacity is 11.5 million tons, down from 12.1 million tons in 1980. It now employs 38,800 workers, down from 63,600 in 1980.

Republic Steel Corporation (merged with LTV Corporation in 1984)

1980:	−1,944	The company operates six steel plants, at Cleveland, Warren, and Canton, Ohio; Buffalo, New York; Chicago, Illinois; and Gadsden, Alabama. In 1980, the firm closes temporarily one of two blast furnaces at Gadsden and one of three blast furnaces at Cleveland.
1981:	+145	No layoffs are announced. The company takes a $36.0 million earnings charge for idle facilities that are not expected to operate in the future.
1982:	−10,634*	The firm announces numerous layoffs, totaling 14,000 (over one-third of the workforce) by the end of the year. Another 4,000 workers are on short work weeks. It closes temporarily some operations at its Chicago plant, idling 2,000 workers, and shuts down two Cleveland bar mills and a coke plant in Buffalo, jointly owned with National Steel. Charges for idle facilities total $25.7 million.
1983:	−5,007	Total layoffs reach 17,000 and are expected to continue. The company takes an earnings charge of $194.1 million for estimated plant shutdown costs, primarily for facilities in Buffalo that it has already closed.
1984:	n.d.	The company merges with LTV Corporation. At this time, the firm has total productive capacity of 10.9 million tons, down from 11.1 million tons in 1980, and 25,250 employees, down from 40,746 in 1980.

National Steel Corporation

1980:	−4,899	The company operates four major steel plants – the Great Lakes plant near Detroit, Weirton in West Virginia, the Midwest plant in Portage, Indiana, and Granite City in Illinois. In 1980, the firm takes a $46.9 million earnings charge for the shutdown of raw materials facilities and the anticipated costs of closing a steel fabricating plant. Layoffs total 2,500 at Great Lakes, 3,000 at Weirton, and 1,600 at Granite City, with further cutbacks likely.

1981:	−880	The firm closes a second blast furnace at Weirton (the first was closed in 1979), leaving two furnaces operating there. It also closes steel finishing facilities at the plant for one week. The company writes off equipment and engineering costs of $16.9 million.
1982:	−6,122*	The firm offers the Weirton plant for sale to its employees, arguing this step is necessary because this operation has the highest employment costs in the steel industry and its 8,400 employees (plus 2,600 on layoff) are unlikely to grant the wage concessions National estimates are required to keep the plant operating. (To keep the USW from organizing Weirton, National has since the 1930s paid workers there about $2.00 more per hour than the average union wage.) The sale of Weirton will reduce steelmaking capacity by about 2.7 million tons. Although no final agreement is reached, the company takes a $286 million charge to 1982 earnings for the anticipated loss on the sale of Weirton. It also takes a $190 million writeoff on raw materials facilities. During the year, the firm announces several cutbacks and layoffs, and plans to close one of two blast furnaces at Granite City for eight weeks.
1983:	−7,154	The company sells the Weirton Steel Division (the transaction actually closes in January 1984), taking an additional $100.0 million loss provision against 1983 earnings. It also takes a $55.6 million writedown on steel facilities. During 1983, the company adopts a holding company organizational structure.
1984:	n.d.	The firm sells a 50% interest in its steel operations to Nippon Steel, recognizing a $16.9 million loss.
1985:	n.d.*	The firm announces plans to eliminate 2,500 hourly and salaried jobs, cutting the work force by 20%. Union officials indicate that they have been promised that the cuts will be made across-the-board in proportion to the current ratios of salaried to hourly workers.
1986:	n.d.	No layoffs or plant closings are announced.
1987:	n.d.	No layoffs or plant closings are announced.
1988:	n.d.	The firm's steelmaking capacity is about 7.5 million tons, down from about 10.1 million tons in 1980. It now employs 19,621 workers, down from 33,856 in 1980.

U.S. Steel Corporation

1980:	−22,482	The firm is the largest domestic integrated steel producer, with 21 major plants in Alabama, Illinois, Indiana, Ohio, Pennsylvania, Texas, and Utah, including the Gary (Indiana) Works; the Lorain (Ohio) Works; the Edgar Thomson, Homestead, Clairton, and National Works near Pittsburgh; South Works at Chicago, Illinois; and the Fairfield (Alabama) Works. Just before the 1980 union negotiations, the firm takes an $808.6 million charge to 1979 earnings to close all or parts of 16 plants and eliminate almost 13,000 jobs. It announces a number of layoffs during the year, the largest of which is 3,000 people in the shutdown of most of its Fairfield operations.
1981:	−7,549	The firm announces a number of one- and two-week closings, but no large layoffs.

1982:	−21,636*	Two plants are merged into an expanded Gary Works. The Geneva (Provo, Utah) and Pittsburg, California plants are combined. The firm consolidates operations at the four Pittsburgh (Pennsylvania) plants, creating the new Mon Valley Works and temporarily suspending some operations. It takes a $123 million earnings charge, primarily employment costs, for the permanent shutdowns.
1983:	−21,265	The company acquires Marathon Oil. It announces a number of permanent (partial and total) shutdowns of operations, for a charge to 1983 earnings of $1.15 billion. The Ensley plant at Fairfield and the Cuyahoga plant at Lorain will be closed. Cutbacks are made at the Gary and Fairfield Works, the Fairless (Pennsylvania) Works, the Geneva Works, and the new Mon Valley Works. These shutdowns and cutbacks will cost 15,400 jobs, of which 10,800 are currently on layoff status, and will reduce steel capacity by 17%. They are announced three days before the fiscal year-end, after the firm's unsuccessful attempt to gain additional USW wage concessions at these marginal operations. The firm announces cutbacks of 4,000 managerial and salaried personnel, above the 8,000 let go since the beginning of 1982.
1984:	−9,969	Bar facilities at Fairless and Duquesne are idled. The Duquesne blast furnace is permanently closed, and the Fairless coke plant idled. Some operations at the Lorain and National Works are combined. The firm agrees to sell its idled Johnstown (Pennsylvania) plant.
1985:	−9,104	Some operations are temporarily idled at Gary and the Mon Valley Works, while others are reactivated at the Fairfield and Texas Works. Earnings reflect a $15 million credit for the 1983 over-accrual of plant shutdown costs in 1983.
1986:	−15,734*	The USW strikes the company for six months, the longest strike in the history of the industry. The firm takes a $1.025 billion earnings charge to cover the indefinite idling of major steel facilities in Utah, Texas, and Pennsylvania. The plants that are idled indefinitely are the Geneva Works in Provo, Utah; the National and Saxonburg plants of the Mon Valley Works; and the hot end and pipe mill at the Texas Works, which together employ 3,700 people and produce 27% of the firm's steel output. The firm emphasizes that these operations are not permanently closed, although this is usually a pre-condition for taking an earnings charge. The company sells a 50% interest in its Pittsburg, California plant to Pohang Iron and Steel Co. of South Korea. It also announces plans to cut another 1,800 administrative jobs, reducing its staff by as much as 15%, and to sell three steel plants, including the Cuyahoga plant (shut down in 1984). The firm adopts a holding company structure, isolating steel and its other operations, in part to demonstrate to the union that steel is subsidized by other operations.
1987:	−10,393	The new labor agreement includes the firm's promise not to shut down permanently the Fairless works during the life of the contract, and to build a continuous caster at the Mon Valley Works. The firm sells the idled Geneva Works and takes an earnings credit of $66 million. The idled Fairless rod mill is sold to Chinese interests, then shipped to China and reassembled there.
1988:	+5,266	The Fairfield Works are restarted, after closing in 1986 for the installation of new machinery. The firm sells the Duquesne, National, and Homestead steel plants, taking an earnings charge of $46 million over the amounts previously provided. The firm's steelmaking capacity is 19.1 million tons, down from about 34.4 million tons in 1980. It now employs 58,778 workers, down from 149,172 in 1980.

Appendix C

Wage and benefit reductions for top management and other salaried personnel in the U.S. steel industry, 1980–1988

This chronology was developed from information in the *Wall Street Journal*. Each entry contains a brief description of the relevant *WSJ* article, often taken verbatim from the article.

Armco Inc.

4/26/82 At the annual meeting, CEO Holiday says financial results this year probably will result in a sharp reduction or elimination of executive bonuses for 1982, and total executive compensation will be cut by 30% to 35%.

7/26/82 The company freezes merit raises and eliminates most bonuses for salaried employees, cutting their total compensation by 15% to 30%.

4/25/83 At the annual meeting, CEO Holiday, apparently anticipating a hostile stockholder reaction to the latest dividend cut (the third in less than a year), tells stockholders that executive compensation for 1982 was down 44% from 1981 levels. He also says the company has suspended all contributions to the employee savings plan for all nonunion, salaried employees, and will try to negotiate a similar arrangement with employees represented by local unions.

Bethlehem Steel Corp.

6/16/82 The firm will reduce salaries of 250 high-level managers by 5% (10% for those who are both officers and directors) beginning July 1. The company also said that it probably won't pay any bonuses this year. The reduction in total managerial compensation will be about 28% for top executives and 14% for other officers. The firm cites severe economic and competitive conditions, but some analysts see the move as a prelude to a dividend cut and as posturing before the steel industry's labor talks start.

7/7/82 The firm announces reductions in its compensation program for 22,000 salaried workers, including 10% cuts in nonofficer directors' fees.

8/10/82 The firm will change its compensation and benefits program for all salaried employees, reducing paid time off, pension, savings, and medical benefits, and instituting an all-merit compensation program.

1/21/83 The firm will again cut the pay of more than 14,000 salaried and nonunion employees, this time by 2.5% to 8%. This action follows a similar move earlier in the week by U.S. Steel, and comes as the industry is entering a third round of bargaining with the USW.

9/30/85 The firm will further reduce wages and benefits for 10,000 salaried workers. The cuts include a 10% salary reduction for upper management, a 5% reduction in the salaries of all other salaried employees, and the elimination of the dental plan. The chairman indicates that similar contributions by hourly employees are needed to help return the firm to profitability and to help provide employment security for as many people as possible.

12/18/85 The firm scales back retirement benefits for 10,000 salaried nonunion employees. The chairman indicates that hourly workers will be expected to make similar concessions in the upcoming union negotiations (opened on 1/22/86).

5/27/86 Article reports white- and blue-collar workers' resentment of CEO Trautlein who, along with other top managers, received a generous salary increase and golden parachute for 1985. White-collar workers have received pay cuts in three of the last four years, and the disclosure of their anger at management apparently contributed to last month's breakdown in union negotiations. Mr. Trautlein announced his retirement (at age 59) last month.

Inland Steel Company

4/29/82 At the annual meeting, the firm announces that it is reducing the pay of about 400 executives and halving the dividend on common stock. It is also freezing the wages of many office employees, and will seek early retirement from others.

12/16/82 The firm is granting smaller bonuses to salaried employees this year, but the USW is angry nonetheless because bonuses will not be eliminated. A company spokesperson justifies the action by noting that salaried workers didn't receive merit increases this year, were denied a cost-of-living adjustment, and had medical benefits reduced. A union spokesperson says the bonuses are 'going to make us take a long, hard look at the company's contract proposals next year (the current contract expires July 31). If they have such a severe cash flow problem, how can they afford to give bonuses to salaried personnel?'

1/31/83 The firm permanently discontinues cost-of-living adjustments and eliminates two paid holidays for 5,600 salaried workers.

LTV Corporation

5/22/85 Article notes that LTV loaned its CEO an interest-free $1 million so he could exercise expiring stock options. The USW is infuriated, given that the loan was made in a year in which LTV had a record $378.2 million loss and that the firm has labeled recent layoffs 'temporary' to avoid having to pay severance benefits to union workers. The amount of the loan would have been sufficient to prevent the loss of more than 600 jobs, according to a union spokesperson.

8/30/85 The firm's troubled steel unit will cut the pay of 8,000 salaried employees, effective September 1. It will also reduce a number of benefits, including vacation days and overtime. One industry source says that LTV has also drawn up a shopping list of concessions it hopes to obtain from unionized workers (union negotiations are expected to begin in December).

1/6/86 Chairman Hay promises further cuts in pay, benefits, and employees at corporate headquarters, and says that he himself has volunteered for a 25% pay cut.

7/18/86 LTV files for bankruptcy protection, and says it will seek further labor cost reductions from employees.

4/13/87 *WSJ* article reveals that the firm paid its top executives almost $1 million in performance-related compensation in 1986, the year it filed for bankruptcy protection and negotiated labor givebacks. One month later, the union rejected the latest labor agreement, in part because it believes managers haven't shouldered as much financial sacrifice as the firm's 23,000 steelworkers.

Republic Steel Corporation (merged with LTV Corporation in 1984)

4/21/82 The firm says it is suspending cost-of-living raises for all salaried employees, effective this month.

7/?/82 The company installs a freeze on merit pay increases (exact date of article unknown, reference is in 8/26/82 article).

8/26/82 The firm will cut salaries 7% to 10% for 5,800 employees. Officers' pay cuts will be 10%, while those for supervisory and other salaried workers will be 7%. Management incentive plans, savings and investment plans, and some paid holidays were previously cut, reducing compensation of management personnel by an average of 18%.

National Steel Corp.

8/21/80 The firm cuts the dividend, noting that revenues are inadequate for the current adverse conditions, despite the firm's cost-cutting actions, which include a reduction in all officers' salaries. National has been especially hard hit because its product mix is tilted heavily toward flat-rolled steel for the automobile industry, which is experiencing a strong downturn.

3/16/82 The firm, expecting a first-quarter loss, is freezing the pay of its 7,000 salaried workers and cutting the pay of senior management by 10% to 25%. It also is discontinuing cost-of-living adjustments for salaried personnel, reducing contributions to the employee stock investment plan, and eliminating some vacation benefits.

10/13/82 The firm will temporarily reduce base salaries by 10% for 6,000 nonunion, salaried employees, and will reinstate them 'as soon as business conditions permit'. The cuts apply to personnel in all operating units and at company headquarters, but not to those at the firm's financial services subsidiary.

3/2/83 The USW ratifies the new labor contract for the steel industry.

3/21/83 The firm cancels the previously-announced 10% salary reduction for nonunion personnel, but will still cut fringe benefits. A company spokesperson says: 'The changes are being made to bring the levels of sacrifice closer to those being made by our hourly workers, and they also expand and make permanent labor cost savings for the present and long term.'

U.S. Steel Corporation

5/20/80 The firm institutes a freeze on managerial (at all ranks) pay less than a month after the completion of union negotiations that granted double-digit wage increases to steelworkers.

3/12/81 The firm seeks shareholder approval to eliminate the effect of such unusual events as plant closings and asset sales on top management's incentive compensation.

6/22/82 The firm will cut compensation benefits of about 27,000 management and salaried employees. Wages are to be cut by at least 5%, with managerial personnel suffering cuts of up to 30% because their incentive compensation will be suspended for 1982. (Incentive pay totaled $3.4 million in 1981.) This announcement is seen as tactical posturing by the steel industry, which is preparing to ask the USW for wage and benefit concessions.

1/17/83 The firm says it expects to report a 'staggering' fourth-quarter loss and will cut the pay of 28,000 management, salaried, and nonunion hourly workers by 5%. The timing of the second nonunion pay cut in seven months is seen as a move to strengthen the industry's position when it again asks the USW next month for steep pay and benefit concessions. Chairman Roderick alleges that only the USW has refused to contribute to the sacrifices made by everyone else in the steel family to ensure the survival of the company.

8/12/86 USX plans to extend (previously announced: date unknown) pay cuts to its Marathon Oil and Texas Oil & Gas units. The cuts, said to be 12% for top officials and 6% for lower-level managers, follow similar cuts for its corporate staff and steelmaking and diversified business units.

3/13/87 USX discloses that top management took pay cuts of about 25% in 1986 as a result of a series of austerity measures adopted when the USW struck the firm in August. However, they also received golden parachutes that would pay as much as three times their salaries plus incentive bonuses if they lost their jobs following a hostile takeover.

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